ScienceDirect

Journal of Manufacturing Processes

Date: 15 August 2026

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Dual-ring spiral laser scanning for welding LPBF Ti600 lattice nodes trajectory-governed melt-pool dynamics and thermal regulation

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10.1016/j.jmapro.2026.05.020

Article

Highlights

- Introduces a geometry-aware dual-ring spiral trajectory decoupling interfacial bridging from lateral fusion expansion for irregular lattice-node interfaces.
- Trajectory-driven energy redistribution promotes partial columnar-to-equiaxed transition by balancing ligament conduction and melt-pool convection.
- Defines a stable process window (~700 W, 125–150 mm/s) that produces symmetric fusion zones with low porosity (~5.20%).
- Links defect suppression and fusion continuity to improved node-level load-bearing performance and fracture transfer to the heat-affected

Outline

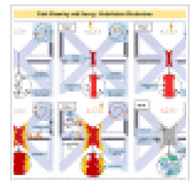
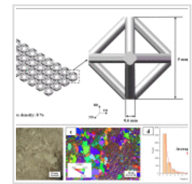
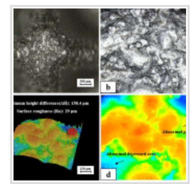
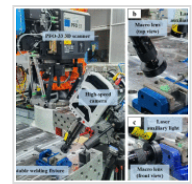
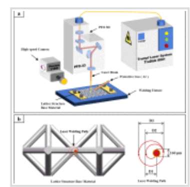
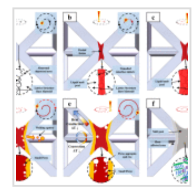
zone.

Abstract

Additively manufactured (AM) Ti600 lattice structures are attractive for high-temperature lightweight components, yet joining lattice nodes by laser welding is hindered by geometry-driven and discontinuous heat extraction, which destabilizes the melt pool and narrows the defect-free process window. This study introduces a geometry-aware dual-ring spiral scanning strategy for scanner-based welding of LPBF Ti600 lattice nodes, where the inner spiral stabilizes initial interfacial bridging and the outer spiral promotes controlled lateral fusion expansion across irregular node interfaces. A process–structure–property framework is established by synchronizing high-speed imaging with fusion-zone morphology, X-ray CT porosity, EBSD reconstruction, and node-level tensile testing over laser powers of 500–900 W and scanning speeds of 75–175 mm/s. An optimal window around 700 W and 125–150 mm/s produces symmetric fusion zones with minimal porosity (~5.20%). The continuously rotating thermal gradient induced by the spiral trajectory disrupts persistent epitaxial growth, promotes a columnar-to-equiaxed transition and reduces texture anisotropy, yielding refined α' martensitic laths of ~3.37–4.38 μm with improved spatial homogeneity. The maximum node-level tensile force reaches 452.4 N at 700 W–150 mm/s, accompanied by a fracture shift from the weld center to the heat-affected zone, indicating that the welded joint is no longer the mechanically limiting region of the lattice node. The results demonstrate that trajectory-level energy delivery provides a mechanism-based route to widen the processing window and reliably join porous metallic architectures.

Graphical abstract

Figures (25)

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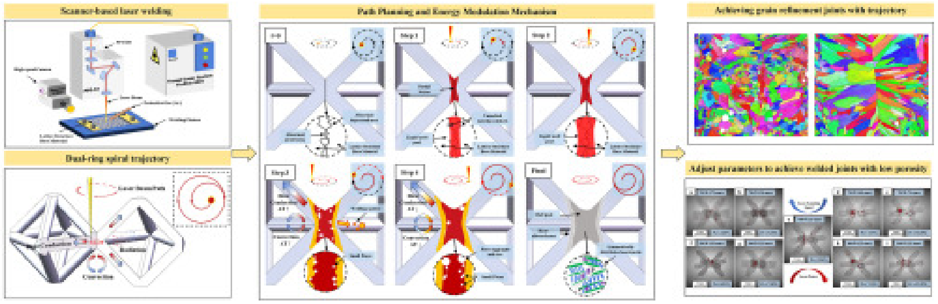
Tables (2)



Keywords

Scanner-based laser welding; Additive manufacturing; Lattice structure joining; Spiral scanning trajectory; Melt-pool dynamics; Thermal regulation

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1. Introduction

Additive manufacturing (AM) enables metallic lattice structures with high stiffness-to-weight ratios, superior energy absorption, and surface-dominated multifunctionality, offering new routes for lightweight aerospace components under thermal load [1], [2]. Recent reviews further highlight that the large surface-area-to-volume ratio of cellular/lattice metals amplifies heat-transfer and functional-integration advantages, motivating geometry-tailored manufacturing and joining strategies for service-critical thermal structures [3]. For high-temperature service, near-α titanium alloys such as Ti600 exhibit improved creep resistance and microstructural stability under elevated-temperature exposure compared with conventional Ti-6Al-4V [4], [5]. However, the limited build volume of laser powder bed fusion (LPBF) systems often necessitates modular fabrication and subsequent joining of lattice sub-components to realize full-scale structures [6]. Reliable node-level joining therefore becomes a critical bottleneck for deploying lattice architectures in engineering-scale assemblies.

Laser welding is attractive for assembling titanium lattices because of its high energy density, localized heat input, and compatibility with automated beam delivery [7]. Nevertheless, welding open-cell lattice nodes differs fundamentally from bulk or sheet structures: heat conduction is constrained by slender ligaments, while multi-facet exposure accelerates heat loss through convection and radiation, leading to discontinuous and anisotropic thermal boundary conditions [8], [9]. This geometry-driven heat extraction destabilizes the melt pool and frequently produces lack-of-fusion, pore entrapment, and irregular fusion morphology that severely degrade joint integrity [9], [10]. In parallel, it has been widely reported that defect sensitivity increases when thermal conditions and melt-pool stability are strongly perturbed, and that porosity and lack-of-fusion remain dominant reliability concerns in load-bearing AM metals [11], [12].

Most defect-mitigation efforts rely on static tuning of nominal process parameters (laser power, travel speed, and focal condition) [7], [9]. For thermally sensitive lattice nodes, however, parameter-only optimization often yields a narrow and unstable processing window: small deviations in local energy balance can trigger either interfacial lack-of-fusion or excessive porosity/shape collapse [10], [13]. This limitation becomes more pronounced in lattice-node welding because the heat source interacts with a discontinuous and curved interface rather than a planar joint line. As a result, the local melt-pool response depends not only on the nominal linear energy density, but also on the instantaneous beam position relative to the struts, node center, and interfacial gaps. Therefore, an effective welding strategy should regulate the spatial sequence of energy deposition, rather than only increasing or decreasing the total heat input. Dynamic beam manipulation—especially beam oscillation—has thus been introduced to redistribute energy, modify melt flow and keyhole behavior, and reduce porosity [14], [15]. Recent in-situ and experimental studies have further shown that beam oscillation can modify keyhole dynamics and improve melt-pool stability during laser processing [16], [17]. Recent titanium-alloy welding studies also demonstrate that oscillation parameters (e.g., frequency and amplitude) can systematically influence porosity and microstructure [13], [18], reinforcing that spatiotemporal energy delivery is a primary control variable rather than a secondary “add-on”.

Beyond simple oscillation, tailored scanning trajectories (e.g., circular or spiral paths) offer additional degrees of freedom by continuously rotating local thermal gradients, regulating dwell/overlap, and thereby altering both melt-pool stability and solidification conditions [19], [20]. Nevertheless, trajectory-controlled welding strategies specifically designed for the discontinuous topology and topographic irregularity of AM lattice nodes remain underexplored, particularly for high-temperature titanium lattices where defect sensitivity and microstructural anisotropy jointly dictate node reliability [10], [21]. Moreover, a mechanistic understanding requires direct temporal evidence of melt-pool response (in-process imaging) in addition to post-mortem morphology and microstructure [22], [23].

To address these gaps, this work proposes a geometry-aware dual-ring spiral scanning strategy for scanner-based welding of LPBF Ti600 lattice nodes. The inner spiral is designed to stabilize initial interfacial bridging via localized preheating, whereas the outer spiral promotes controlled lateral fusion expansion to achieve symmetric fusion coverage across irregular node interfaces. High-speed imaging combined with confocal surface profiling, X-ray computed tomography, and EBSD reconstruction is employed to quantify the coupling between trajectory-controlled heat input and lattice-specific heat dissipation. This integrated characterization establishes a process–structure–property framework linking melt-pool dynamics to defect suppression, solidification-mode transition, and node-level mechanical performance.

2. Experimental procedures and details

2.1. Material, structure and interface morphology

The substrate material for this study is Ti600, a Chinese-developed near-α high-temperature titanium alloy specifically engineered for aerospace propulsion components. Characterized by its exceptional structural stability and oxidation resistance at temperatures up to 600°C, this alloy is a primary candidate for replacing conventional Ti-6Al-4V in high-thermal-load environments. The chemical composition of the LPBF-fabricated Ti600 used in these experiments is summarized in Table 1.

Table 1. The chemical compositions of Ti600.

Materials	Nominal chemical compositions(wt%)							
	Al	Sn	Zr	Mo	Si	Y	Fe	Ti
Ti600	6.95	2.43	4.4	0.345	0.229	0.152	0.031	Bal.

The lattice structure features an octahedral unit cell configuration, synthesized via the Laser Powder Bed Fusion (LPBF) process. As illustrated in Fig. 1a, each unit cell measures 5 mm in height with a nominal strut diameter of 0.6 mm, achieving a relative density of approximately 8%. Microstructural analysis of the as-printed base material (BM) via optical microscopy (Fig. 1b) and EBSD (Fig. 1c) reveals a matrix dominated by fine, acicular α' martensite, reflecting the rapid solidification characteristic of the LPBF process. Grain boundary statistics indicate a high-angle grain boundary (HAGB >15°) fraction of approximately 96.2%, suggesting a complete and uniform martensitic transformation from the prior-β phase. According to the grain size distribution shown in Fig. 1d, the BM exhibits an average equivalent circle diameter of 7.74 μm, with most measured grains concentrated in the fine-size range and a long-tailed distribution toward larger grain sizes. This indicates a relatively fine martensitic microstructure in the as-built lattice strut. No significant crystallographic texture is observed in the as-built base material. No significant crystallographic texture is observed in the as-built base material. The initial porosity of the lattice struts was measured at 1.20% via CT scanning, indicating a relatively dense and defect-controlled initial condition for subsequent welding experiments. For clarity, a unified coordinate system is adopted throughout this study. The RD is defined as the laser incident (beam) direction, which remains consistent for both the LPBF fabrication and the subsequent laser welding process. The TD and ND are defined as two mutually orthogonal directions lying in the plane normal to RD; accordingly, the laser focal plane is defined as the TD–ND plane. The laser scanning direction lies within this focal plane and is specified in Fig. 1.

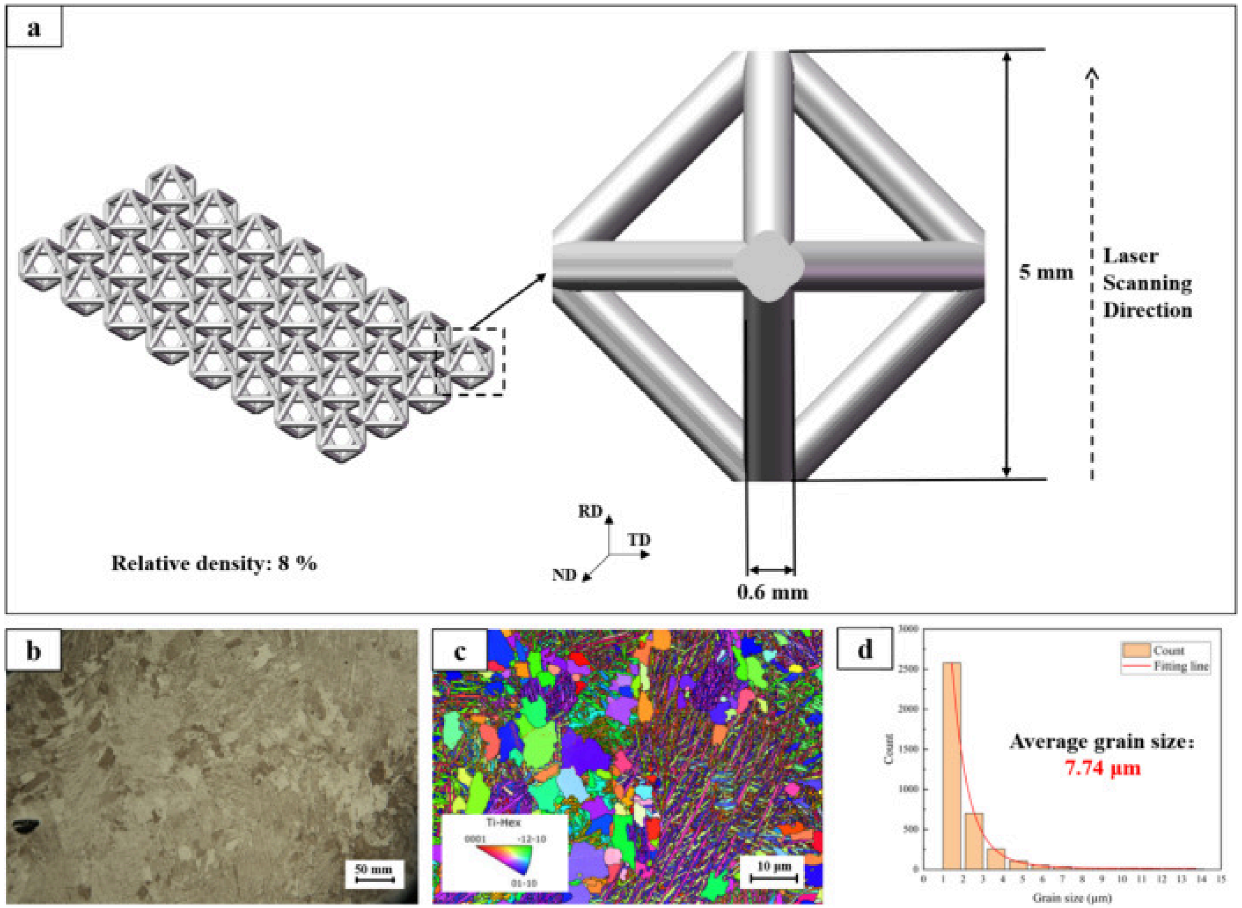

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Fig. 1. Welding base material specimen and metallographic. (a) Schematic of the lattice structure manufactured by LPBF additive manufacturing, and dimensions of a base unit of the lattice structure, (b) metallographic photo of the as-manufactured lattice structure, (c) EBSD maps showing Band Contrast (BC), Inverse Pole Figure (IPF), and Grain Boundary (GB) distributions of the base material, (d) Grain size distribution (equivalent circle diameter) obtained of the base material.

The geometric integrity of the mating interface is primarily affected by surface irregularities originating from the LPBF process and post-processing operations. As illustrated in the optical micrograph (Fig. 2a) and the corresponding confocal laser scanning microscopy (CLSM) image (Fig. 2b), the interface exhibits pronounced topographic fluctuations and micro-scale irregularities. The CLSM measurements were performed using a Keyence VK-X200 laser confocal scanning microscope, with a vertical resolution of approximately 1 nm and a minimum step size of 3.8 nm.

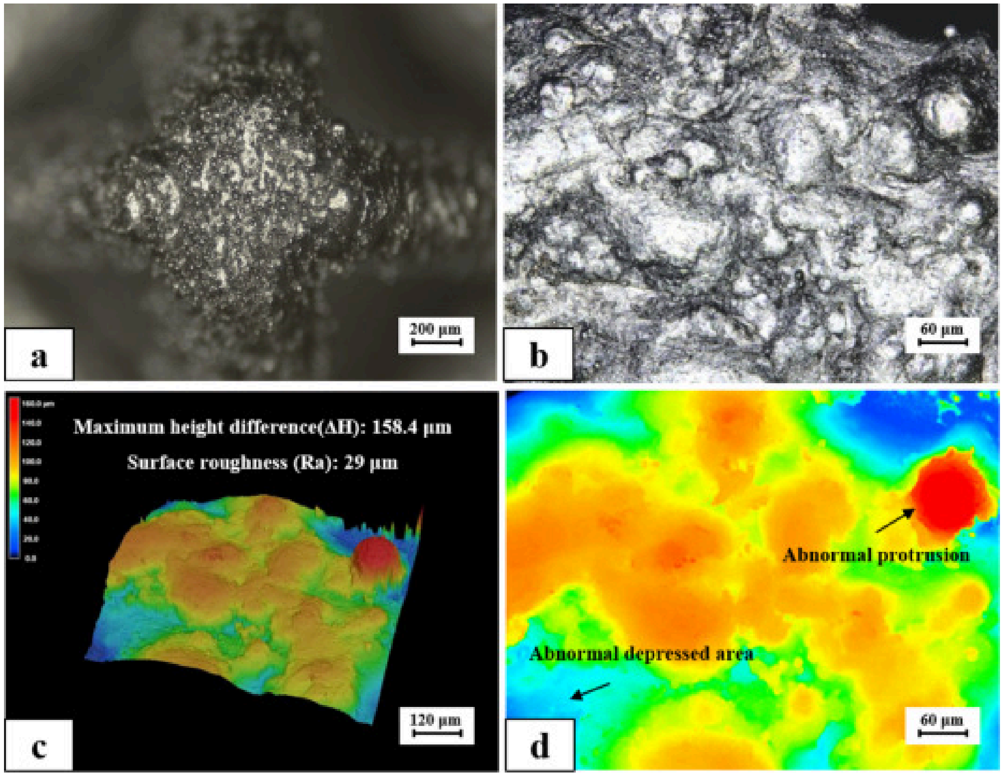

Fig. 2. Surface morphology and geometric defects of the node mating interface. (a) Optical micrograph of the lattice node interface; (b) CLSM image acquired using a Keyence VK-X200 laser confocal scanning microscope; (c) 3D topographic height map showing the measured surface roughness and maximum height difference; (d) color-coded height distribution map identifying abnormal protrusions and depressed areas.

Quantitative characterization through 3D height mapping (Fig. 2c) reveals a substantial surface roughness (Ra) of 29 μm and a maximum height difference (ΔH) of 158.4 μm. The color-coded height map (Fig. 2d) further distinguishes specific “abnormal protrusions” and “abnormal depressed areas” across the contact surface. These inherent defects introduce non-uniform micro-gaps at the joint interface, which can act as sources of perturbation for the laser-induced melt pool during welding. These discontinuities may introduce localized thermal perturbations during laser irradiation, potentially resulting in unstable melt pool behavior, incomplete fusion, or spatter formation, which could adversely affect joint reliability.

Therefore, a dual-ring spiral scanning strategy was adopted to accommodate the geometric irregularities at the interface. The primary scanning pass functions as a "geometry-aware" buffer, utilizing localized preheating and bridging to mitigate existing topographic irregularities. Subsequently, the secondary pass achieves stable, deep-penetration fusion within a homogenized thermal field. This strategy is critical for overcoming the inherent surface limitations of additively manufactured lattice nodes and ensuring high-performance joint integrity.

2.2. Experimental setups and procedure

The experimental configuration for laser welding is shown in Fig. 3. The system consists of a TruDisk 8001 disk laser (maximum output power: 8 kW; fiber core diameter: 50 μm; beam expansion ratio: 3.2; wavelength: 1030 nm) integrated with a PFO-33 3D galvanometer scanner mounted on a KUKA industrial robot. To ensure accurate positioning and mechanical stability during welding of discrete lattice nodes, a customized adjustable fixture was designed and employed (Fig. 3a).

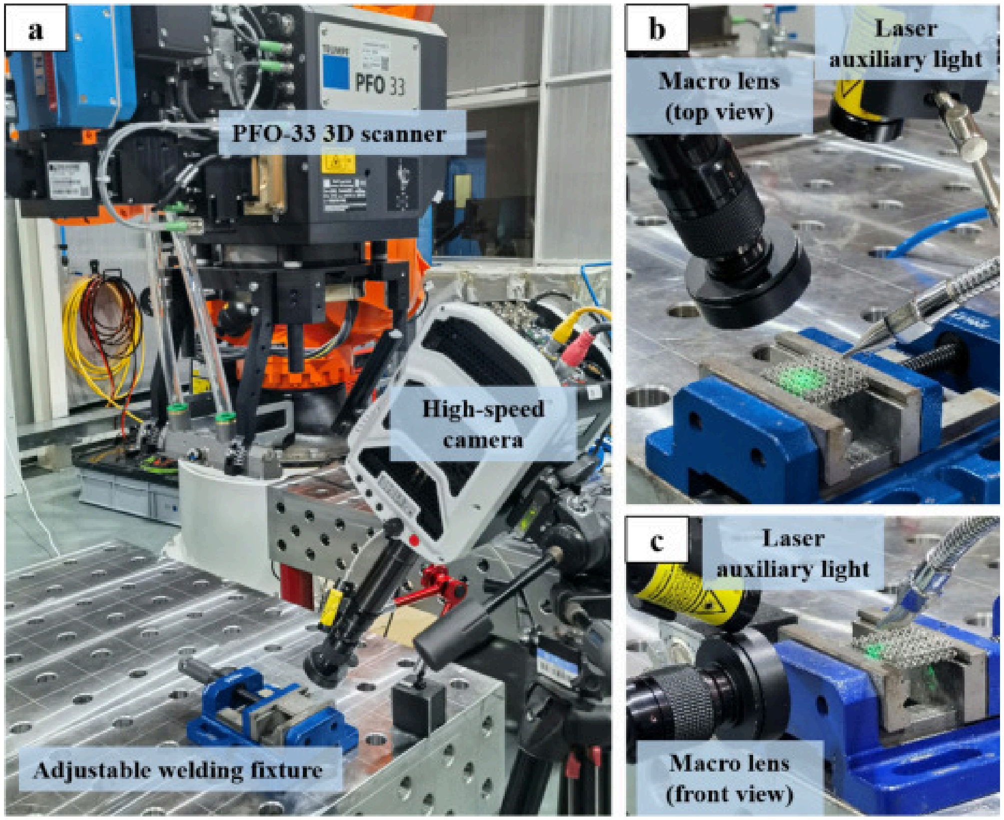

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Fig. 3. Experimental setup for laser welding of AM lattice structures. (a) Integral view of the robotic laser welding system and high-speed monitoring assembly, (b) quasi-top view showing the macro lens orientation and laser auxiliary light, (c) front view of the welding fixture and observation angle.

An important feature of the system is the real-time observation capability. A Phantom V1840 high-speed camera, equipped with a high-magnification macro lens and a laser auxiliary illumination source, was used to record the transient melt pool behavior. As shown in the quasi-top view (Fig. 3b) and front view (Fig. 3c), the dual-view monitoring enables detailed characterization of melt pool morphology, surface oscillation, spatter dynamics, and melt pool stability.

The welding strategy adopted in this study is schematically illustrated in Fig. 4. It should be noted that this is a single-beam scanner-based laser welding strategy, rather than a multi-beam or simultaneous multi-spot process. The dual-ring spiral path was planned as a predefined galvanometer scanning trajectory and executed by the PFO-33 scanner during the post-LPBF welding stage. During welding, the robot and specimen remained stationary, while the single laser beam was rapidly deflected by the galvanometer to follow the programmed spiral trajectory under zero-defocus conditions. This method enables precise and repeatable localized energy delivery at lattice node intersections without mechanical stage motion. By adjusting laser power and scanning speed, the local energy density and interaction time can be effectively controlled, allowing regulation of melt pool size and penetration depth under severe geometric constraints. Such control is essential for stabilizing the melt pool on slender struts, mitigating lack-of-fusion or structural collapse, and tailoring the solidification behavior.

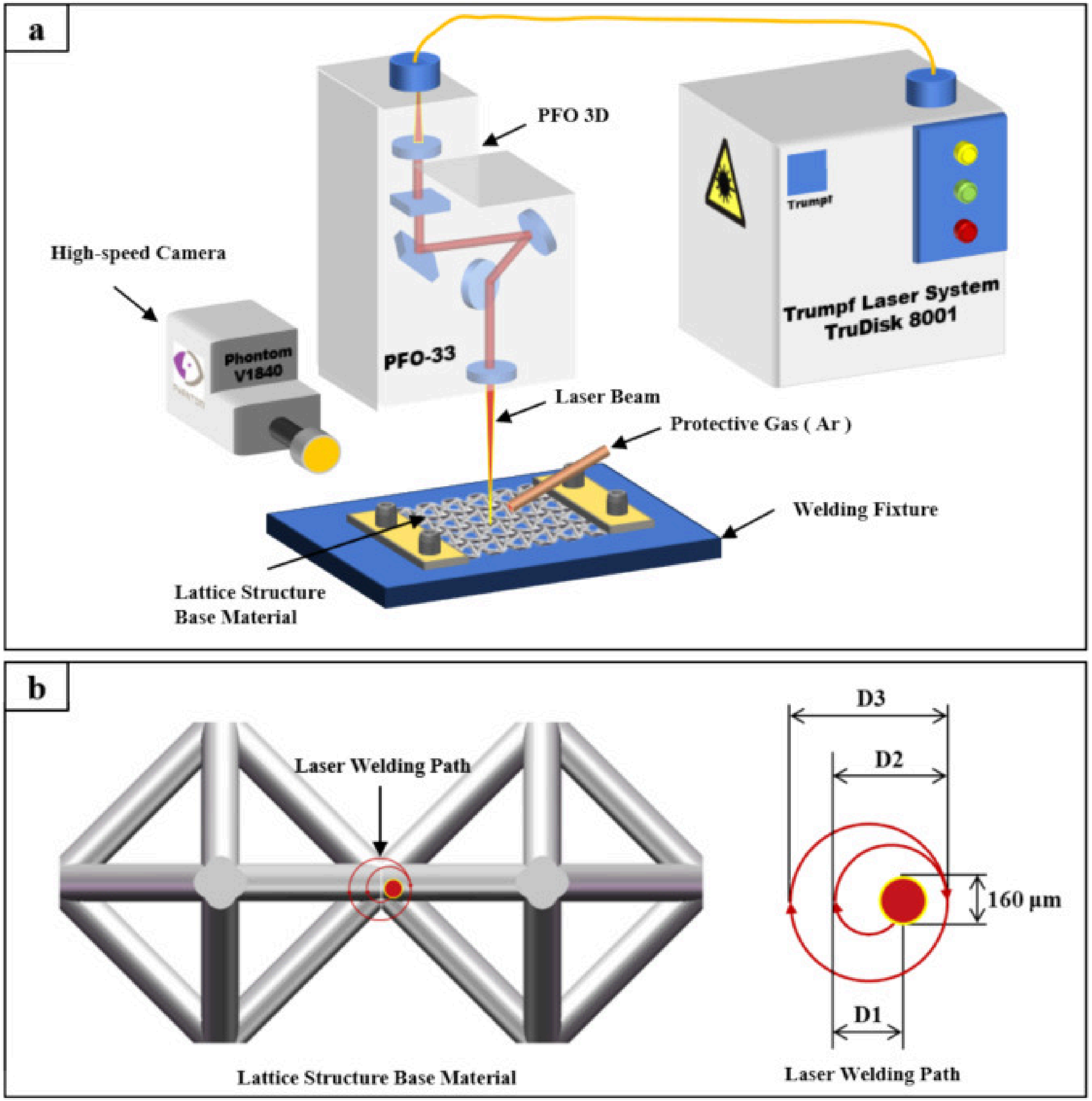

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Fig. 4. Schematic of the single-beam dual-ring spiral scanning principle and path design. (a) Principle schematic of the scanner-based post-LPBF laser welding system, in which a single laser beam is deflected by the galvanometer scanner; (b) top-down view of the predefined spiral trajectory with specific diameters and path dimensions.

The geometry-aware scanning strategy consists of concentric multi-loop spiral motions, as illustrated in Fig. 4b. The specific dimensions of the spiral trajectory were determined by considering the characteristic node scale, the nominal strut diameter, the interfacial height fluctuation, and the laser spot diameter. For the LPBF Ti600 lattice used in this study, the strut diameter is approximately 0.6 mm, while the measured maximum height difference at the mating interface reaches 158.4 μm. Therefore, the inner loops were designed with D1 = 0.24 mm and D2 = 0.40 mm to concentrate the initial energy input near the node intersection, promote localized preheating, and rapidly establish a stable liquid bridge across the irregular interface. This inner-ring design avoids excessive early melting of the peripheral ligaments while ensuring that the central micro-gaps can be wetted and sealed.

The outer loops were designed with D3 = D4 = 0.56 mm, which is close to the nominal strut diameter and thus provides sufficient lateral coverage of the node interface without extending the fusion zone excessively beyond the load-bearing ligament region. In this way, the outer spiral expands the melt pool after the initial bridge has formed, enlarges the effective bonding area, and redistributes heat laterally to suppress local overheating at the weld center. The radial spacing between adjacent loops was set to approximately 160 μm, matching the laser spot diameter under zero-defocus conditions. This spacing maintains continuous energy overlap between neighboring loops while avoiding repeated excessive irradiation of the same central region. Thus, the selected D1–D4 dimensions and loop spacing were not arbitrary, but were designed to decouple initial interfacial bridging from subsequent peripheral coverage, which is the key physical rationale of the proposed dual-ring spiral trajectory.

It should be noted that this trajectory was selected as a representative configuration to verify the proposed inner-bridging/outer-coverage principle, rather than to exhaustively optimize all possible spiral geometries. The objective of this work is to establish whether staged spatial energy delivery can stabilize melt-pool formation at geometrically irregular lattice nodes. Further optimization of the inner- and outer-loop dimensions, loop number, and overlap ratio may be required when the strategy is extended to lattice structures with different strut diameters, node geometries, or material systems.

Prior to welding, the specimens were ultrasonically cleaned in acetone to remove surface contaminants. All experiments were conducted under high-purity argon shielding (99.99%) with a flow rate of 20L/min to minimize oxidation. During welding, the robotic arm remained stationary, and the beam trajectory was controlled exclusively by the galvanometer scanner. Therefore, the spiral path was implemented after LPBF fabrication as a post-processing joining operation, not during powder-bed deposition. The strategy is not specific to the present robot–scanner configuration; in principle, it can be implemented on other scanner-based laser welding or laser processing systems that provide programmable beam positioning, sufficient positioning accuracy, and an appropriate laser power range. However, it should not be interpreted as a path that can be directly applied to any LPBF machine unless that machine is equipped with suitable post-process or in-situ beam-scanning capability for localized node welding.

To systematically investigate the influence of energy input on melt pool stability under the geometric constraints of lattice nodes, a full-factorial experimental design was adopted. Due to the limited thermal mass and complex heat dissipation characteristics of the slender strut network, the process window for stable welding is relatively narrow, where insufficient energy leads to lack of fusion, while excessive energy results in structural collapse or keyhole-induced porosity. Preliminary screening experiments were first conducted to identify a feasible processing range. Based on these results and previous single-ring scanning studies, laser power (P) and scanning speed (v) were selected as the primary independent variables, as they determine the nominal linear energy density (E=P/v) and the interaction time at the lattice node.

A 5×5 parameter matrix was then established, with laser power varying from 500 to 900W (in increments of 100W) and scanning speed ranging from 75 to 175mm/s (in increments of 25mm/s), as summarized in Table 2. This matrix generates 25 distinct processing conditions covering the transition from an energy-deficient lack-of-fusion regime to an over-melting excessive-porosity regime. By isolating the effects of power and speed under a fixed dual-ring spiral trajectory (D1–D4), the study aims to construct a process–melt pool stability–joint quality map for Ti600 lattice welding. Unless otherwise specified, all experiments were performed under zero defocus conditions, resulting in a constant laser spot diameter of approximately 160μm.

Table 2. Laser welding process parameters.

No.	Laser power (W)	Scanning speed(mm/s)	No.	Laser power (W)	Scanning speed(mm/s)
1	500	175	14	800	125
2	600	175	15	900	125
3	700	175	16	500	100
4	800	175	17	600	100
5	900	175	18	700	100
6	500	150	19	800	100
7	600	150	20	900	100
8	700	150	21	500	75
9	800	150	22	600	75
10	900	150	23	700	75
11	500	125	24	800	75
12	600	125	25	900	75

No.	Laser power (W)	Scanning speed(mm/s)	No.	Laser power (W)	Scanning speed(mm/s)
13	700	125			

2.3. Methods for microstructure and property characterization

Welded specimens were extracted from the lattice node region and sectioned along two orientations, parallel and perpendicular to the laser focal plane, followed by standard metallographic preparation and etching in Keller's reagent for 20–30 s. The macro- and microstructures were examined using a Leica optical microscope (OM). Surface morphology and interface topography were characterized using a Keyence VK-X200 laser confocal scanning microscope (CLSM) with a vertical resolution of approximately 1 nm and a minimum step size of 3.8 nm. Crystallographic characterization was performed by Electron Backscatter Diffraction (EBSD) on ion-polished surfaces using a Tescan MIRA3 field-emission scanning electron microscope (FE-SEM) operated at 20 kV with a step size of 0.15 μm. Transient melt pool behavior during welding was recorded using a Phantom V1840 high-speed camera, with a minimum exposure time of 1 μs, enabling the observation of melt pool evolution, surface oscillation, and spatter dynamics. Quasi-static tensile tests were conducted using a SUNS UTM5504X universal testing machine at a crosshead speed of 0.5 mm/min. For each processing condition, three specimens were tested to ensure statistical reliability, and the average tensile strength was reported.

3. Results and discussion

3.1. Dual-ring spiral path principle and melt pool dynamics

3.1.1. Path planning and energy modulation mechanism

The spatiotemporal energy modulation mechanism of the dual-ring spiral scanning strategy is schematically illustrated in Fig. 5. By separating the initial interfacial wetting from subsequent melt pool expansion, this geometry-aware path planning enables stable fusion at lattice nodes despite the severe geometric constraints of the Ti600 strut network. Prior to laser irradiation, the node mating interface exhibits pronounced topographic irregularities, including abnormal protrusions and locally depressed regions introduced during the LPBF process, which create non-uniform micro-gaps and discontinuous heat conduction paths. During the initial half-loop of the inner spiral, localized laser irradiation induces partial melting at the node intersection and rapidly establishes a narrow liquid bridge through surface-tension-driven wetting. As the second half-loop proceeds, this liquid bridge evolves into a continuous preliminary melt pool, while the surrounding material undergoes gradual preheating, reducing the local temperature gradient and suppressing abrupt collapse of the slender struts. When the scanning trajectory transitions to the outer spiral loops, the melt pool undergoes significant lateral expansion and increased penetration depth, accompanied by intensified interaction between axial heat conduction along the struts and convective flow within the molten metal. Although transient vaporization may induce sporadic spatter or small gas pores at this stage, the continuous spiral motion effectively prevents excessive heat accumulation at the node center, which is commonly associated with keyhole instability in single-ring or static scans. With further spiral progression, the cumulative energy input is redistributed over a broader area. This spatial redistribution is expected to reduce localized overheating at the node center and to moderate the dominance of a single directional heat-flow path, thereby weakening violent melt-pool fluctuations and allowing entrapped vapor bubbles to aggregate and rise toward the molten surface during the extended molten lifetime. Upon completion of the spiral trajectory, the melt pool solidifies under a relatively controlled cooling condition, producing a symmetrically shaped fusion zone with a uniform solidification pattern. Compared with the narrow and defect-prone process window of single-ring scanning, the dual-ring spiral strategy accommodates the inherent surface irregularities of LPBF lattice nodes while maintaining sufficient fusion coverage and a smooth weld crown, thereby providing a stable foundation for subsequent defect suppression and microstructural refinement. In this schematic, the color coding in the heating stages is used to distinguish the transient molten region, thermally affected region, and surrounding base material. After cooling, the thermal partition is no longer explicitly distinguished, and the final weld morphology is therefore represented mainly by the weld-boundary contour.

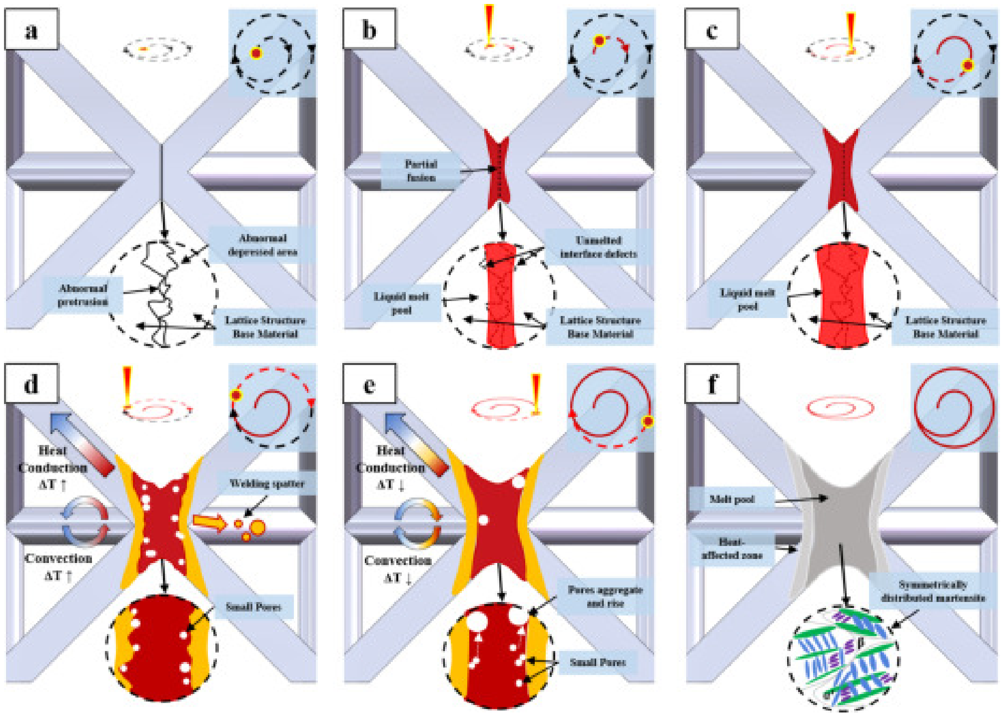

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Fig. 5. Schematic illustration of melt pool evolution during dual-ring spiral laser welding of AM Ti600 lattice nodes. (a) Initial surface condition prior to laser irradiation, (b) Formation of a liquid bridge during the first half-loop of the inner spiral, (c) Development of a continuous melt pool during the second half-loop of the inner spiral, (d) Melt pool expansion during outer-ring spiral scanning, (e) Thermal homogenization and pore evolution during continued spiral motion, (f) Final weld morphology after solidification and cooling.

3.1.2. Transient melt pool evolution and stability based on high-speed observation

The theoretical spatiotemporal energy modulation mechanism proposed in Section 3.1.1 is corroborated by the transient melt pool dynamics captured via multi-angle high-speed imaging (Fig. 6 and Fig. 7). These temporal sequences resolve the evolution of the molten pool at lattice nodes across five representative parameter sets, where frames 1–6 correspond to the characteristic stages of the dual-ring spiral cycle (t=0→t=T). The front-view observations (Fig. 6) specifically highlight the vertical progression from interfacial bridging to stabilization, while the top-view images (Fig. 7) resolve the lateral spreading symmetry and boundary dynamics.

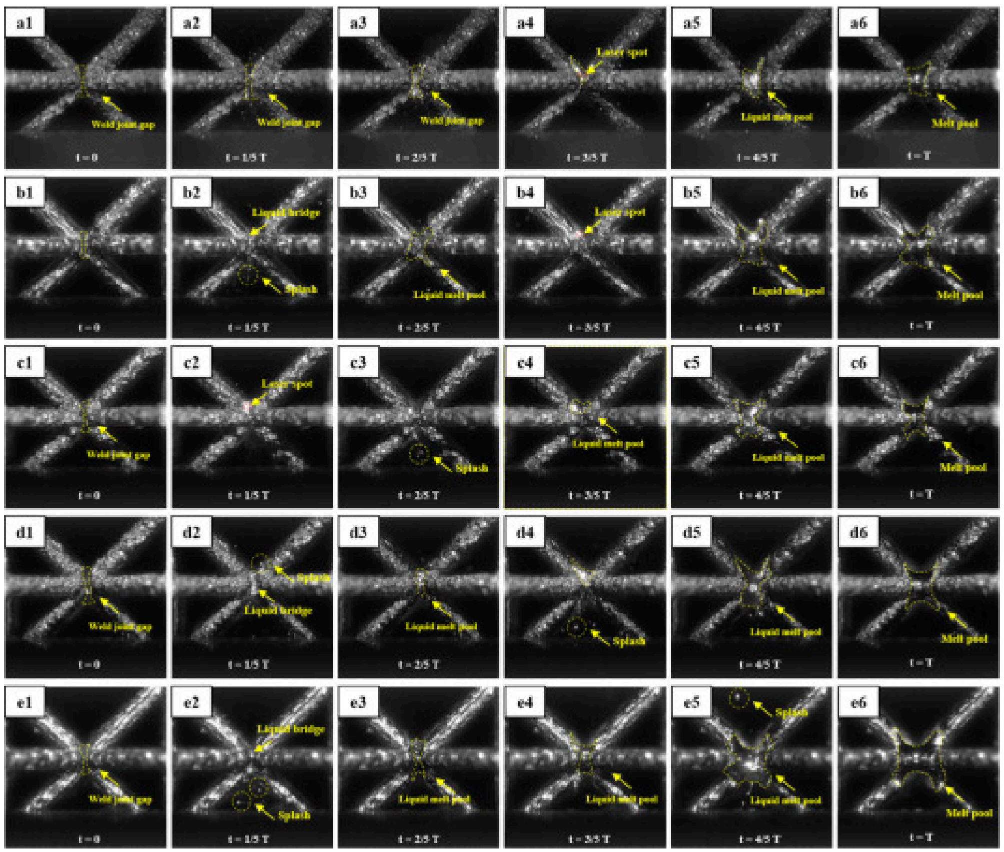

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Fig. 6. Front-view high-speed observation of transient melt pool evolution under different process parameters. (a) 700W–175mm/s, (b) 500W–125mm/s, (c) 700W–125mm/s, (d) 900W–125mm/s, and (e) 700W–75mm/s.

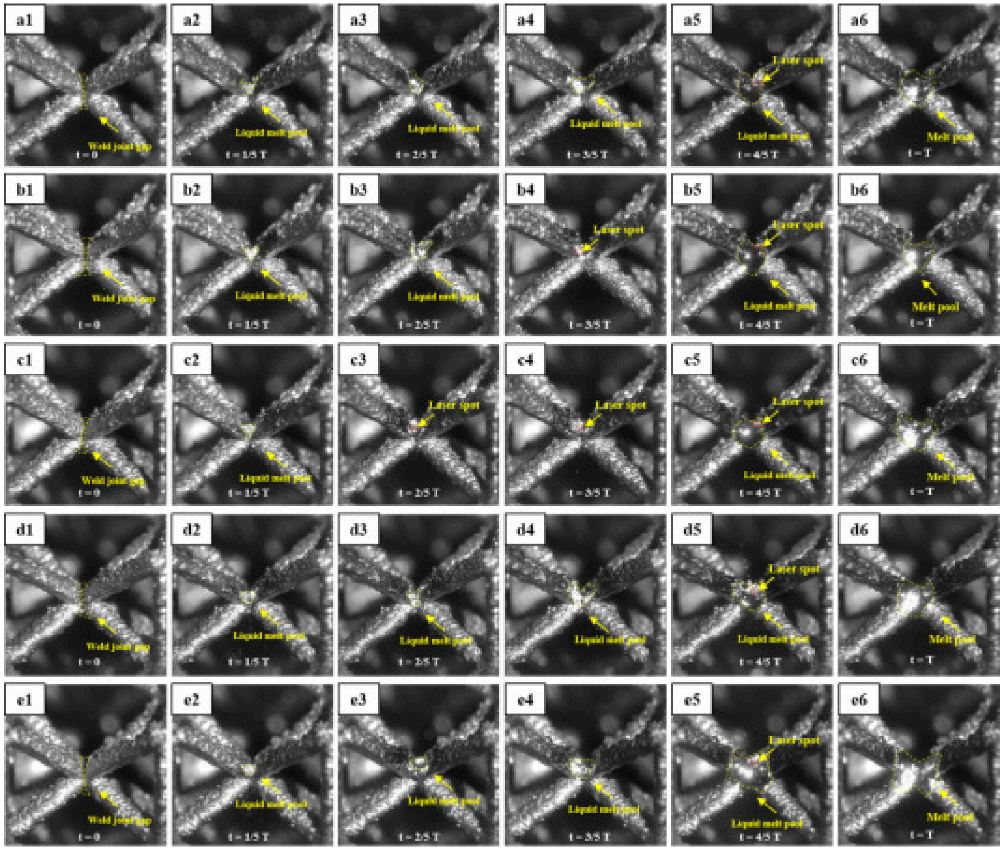

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Fig. 7. Top-view high-speed observation of transient melt pool evolution under different process parameters. (a) 700W–175mm/s, (b) 500W–125mm/s, (c) 700W–125mm/s, (d) 900W–125mm/s, and (e) 700W–75mm/s.

At the onset (t=0), a distinct joint gap is visible at the node intersection across all conditions (Fig. 6a1–e1). This gap is associated with the initial mating condition of the LPBF lattice nodes, including as-built surface roughness, local height fluctuation, positioning tolerance, and incomplete contact between curved strut surfaces. Therefore, the visible joint line is not caused by a specific welding parameter, but represents the inherent interfacial discontinuity that the inner spiral is designed to bridge during the early welding stage. Upon laser irradiation during the inner-ring phase (t=1/5T), localized melting initiates a liquid bridge driven by surface-tension-assisted wetting. This early-stage behavior is highly sensitive to the linear energy density. Under low thermal input, exemplified by Group b (500W, 125mm/s) and Group a (700W, 175mm/s), the liquid bridge remains narrow and sluggish. This indicates that the energy is insufficient to rapidly overcome the geometry-induced heat dissipation of the slender struts, leading to a high risk of incomplete fusion at the interface corners.

In contrast, high-heat-input conditions—Group d (900W, 125mm/s) and Group e (700W, 75mm/s)—establish a robust liquid bridge almost instantaneously. The higher energy density facilitates rapid interfacial sealing, effectively “bridging” the topographic irregularities identified in Section 2.2. The influence of process parameters becomes most critical as the scanning trajectory transitions to the outer spiral loops (t=3/5T→t=4/5T). During this expansion phase, distinct stability regimes emerge: (1) Optimal Stability Window: For Group c (700W, 125mm/s), the melt pool exhibits an exceptionally stable evolution. It maintains smooth, symmetric boundaries with minimal spatter ejection in both front and top views (Fig. 6c4, Fig. 7c4). This suggests that the dual-ring spiral path successfully homogenizes the thermal field, balancing the axial heat conduction along the struts with the convective flow within the pool, thereby preventing the keyhole instability commonly seen in single-track scans. (2) Instability and Over-melting: Excessive energy input triggers pronounced perturbations. In Group d, the high laser power induces intense localized vaporization, where the resulting vapor recoil pressure overcomes surface tension, leading to violent oscillations and large-scale spatter ejection (Fig. 6d4). Similarly, the reduced scanning speed in Group e promotes excessive thermal accumulation. The heat input exceeds the structure's limited dissipation capacity, triggering vigorous convection and frequent splashing that compromise the geometric accuracy of the node.

Upon completion of the spiral trajectory (t=T), the final morphology reflects the cumulative effect of the spatiotemporal energy delivery. Group c yields a continuous, symmetric melt pool with a well-formed weld crown and sufficient penetration. In contrast, Group a shows residual interfacial gaps (incomplete fusion), while Group e exhibits structural sinking due to a prolonged liquid lifetime. Overall, these results confirm that the dual-ring strategy stabilizes the fusion process through a staged thermal cycle: initial liquid bridging, controlled lateral expansion, and homogenized cooling. The melt pool behavior observed for Group c provides a favorable physical foundation for the defect suppression and microstructural refinement analyzed in the subsequent sections.

3.1.3. Solidification and cooling characteristics

The solidification trajectory of the melt pool, following the completion of the dual-ring spiral scanning cycle (t=T), is governed by the geometry-induced heat dissipation characteristics of the Ti600 lattice structure. As revealed by the high-speed imaging sequences in Fig. 8, the transition from a fully liquid state (Fig. 8a) to a solidified joint proceeds under a pronounced spatial thermal gradient. Immediately after laser shut-off, solidification is initiated preferentially at the strut–melt pool interfaces (Fig. 8b), where the slender lattice struts serve as the primary axial heat sinks along the lattice structure. Due to the high surface-area-to-volume ratio and multi-directional connectivity of the struts, heat is rapidly extracted along the axial directions of the ligaments, inducing directional cooling and an inward propagation of the solid–liquid interface. As solidification advances toward the node center (Fig. 8c), the remaining molten region appears to maintain a longer liquid lifetime. This delayed solidification suggests that heat extraction in the pool center is less directly governed by axial conduction through the struts than that near the fusion boundary, although the local temperature gradient and cooling rate were not directly measured in the present study. Final solidification (Fig. 8d) thus yields a weld characterized by a spatially non-uniform cooling history. Therefore, the cooling interpretation in this section is based on high-speed observation of the solidification sequence and post-solidification morphology, rather than on direct thermal-field measurement or numerical simulation.

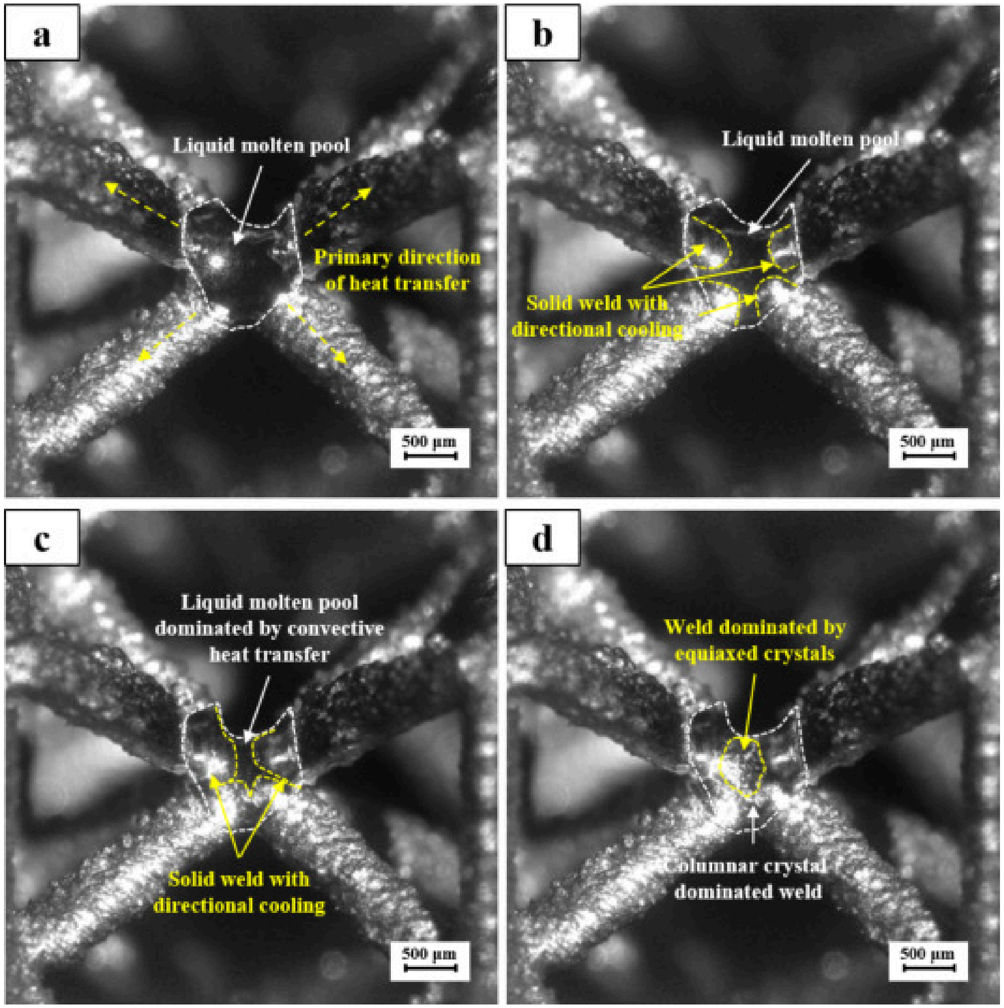

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Fig. 8. High-speed observations of the transient solidification and cooling process after dual-ring spiral welding. (a) Fully liquid melt pool at t=T, (b) initiation of solidification

and directional cooling at the strut interfaces, (c) propagation of the solidification front toward the node center, (d) fully solidified joint showing the final weld morphology.

The post-solidification microstructures observed via SEM (Fig. 9) are consistent with the solidification sequence inferred from the high-speed images. The macroscopic weld morphology (Fig. 9a) exhibits a heterogeneous grain distribution that can be divided into two characteristic zones. Near the fusion boundaries adjacent to the lattice struts (Fig. 9c), the microstructure is dominated by columnar crystals that grow epitaxially from the base material. Their growth directions are preferentially aligned with the primary heat-transfer paths along the struts, suggesting that directional conductive heat extraction plays a dominant role near the fusion boundary.

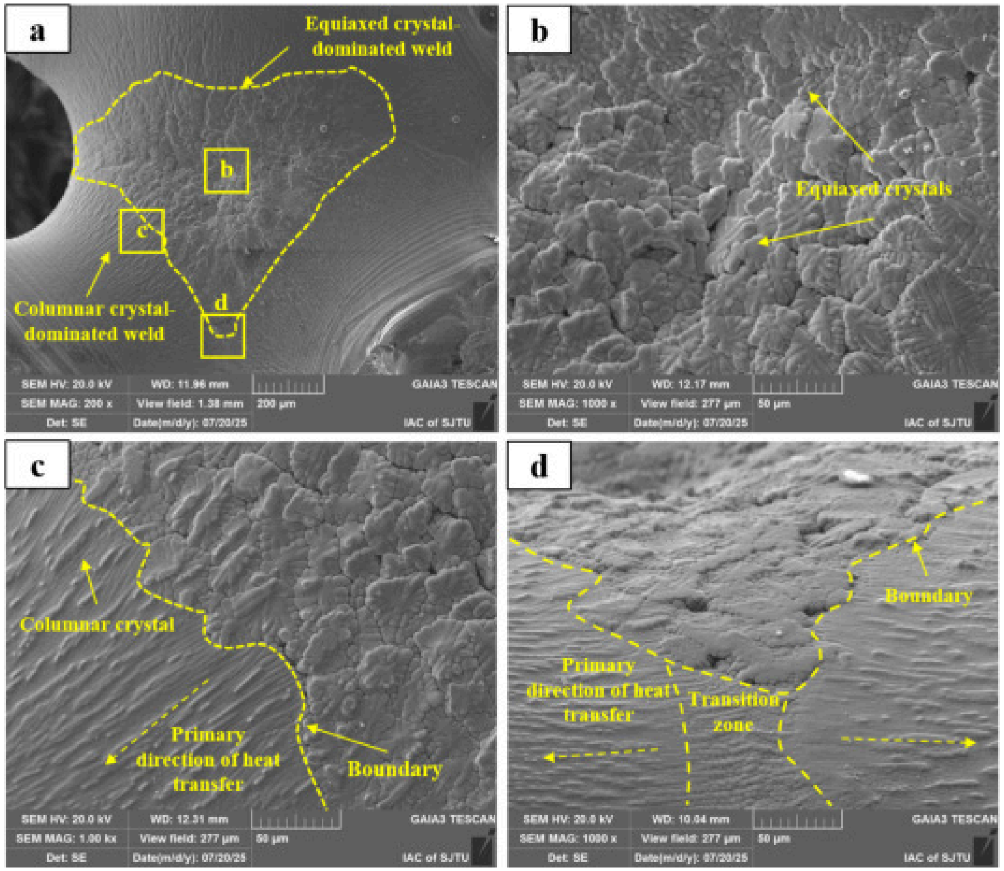

Fig. 9. SEM characterization of the grain morphology and solidification zones in the Ti600 node joint. (a) Macro-morphology of the solidified weld, (b) equiaxed crystal-dominated region in the central fusion zone, (c) columnar crystal-dominated region at the fusion boundary showing directional growth, (d) transition zone between columnar and equiaxed crystals highlighting the solidification boundary.

In contrast, the central region of the weld (Fig. 9b) is characterized by a predominance of equiaxed crystals. Since the local temperature gradient, cooling rate, and G/R ratio were not directly quantified in this study, the observed columnar-to-equiaxed transition should be interpreted as a microstructural indication rather than a direct measurement of thermal-field evolution. Based on classical solidification theory, the formation of equiaxed grains in the weld center is consistent with a reduced dominance of directional temperature gradients and/or an increased effective nucleation tendency in the remaining liquid pool. The characteristic length l and thermal diffusivity a may affect the thermal diffusion time scale of the lattice node, but they do not directly determine G/R without transient thermal-field measurement or simulation. Therefore, no quantitative proportional relationship between G/R and l/a is assumed in this work. A well-defined transition zone is observed between the columnar region near the fusion boundary and the equiaxed region in the weld center (Fig. 9d), corresponding to a columnar-to-equiaxed transition (CET). This transition suggests a dynamic competition between geometry-driven directional heat conduction through the lattice struts and trajectory-induced redistribution of the molten pool, rather than proving a specific quantitative G/R value. Consequently, the weld exhibits a spatially differentiated grain structure comprising peripheral columnar growth and central equiaxed refinement, which provides a microstructural basis for the joint integrity and mechanical performance discussed in subsequent sections.

3.2. Macro-morphology and defect distribution

The transient melt pool stability and solidification behavior discussed in Section 3.1 are fundamentally reflected in the final macro-morphology and internal soundness of the Ti600 joints. This section systematically evaluates the influence of laser power (P) and scanning speed (v) on the joint formation, focusing on the competitive evolution of geometric dimensions and defect distribution across the established processing window. By correlating the energy input parameters with the resulting fusion characteristics, the physical mechanisms governing the transition from insufficient fusion to over-melted regimes are elucidated.

3.2.1. Macro-morphology and geometric consistency

The macroscopic integrity and geometric consistency of the welded Ti600 lattice nodes were systematically quantified through two orthogonal observation planes, as schematically illustrated in Fig. 10. This dual-perspective approach allows for the simultaneous evaluation of the effective fusion diameter (D) on the node surface and the penetration depth (H) into the vertical ligaments. The morphological evolution across the 5×5 parameter matrix is presented in Fig. 11 (top view) and Fig. 12 (front view).

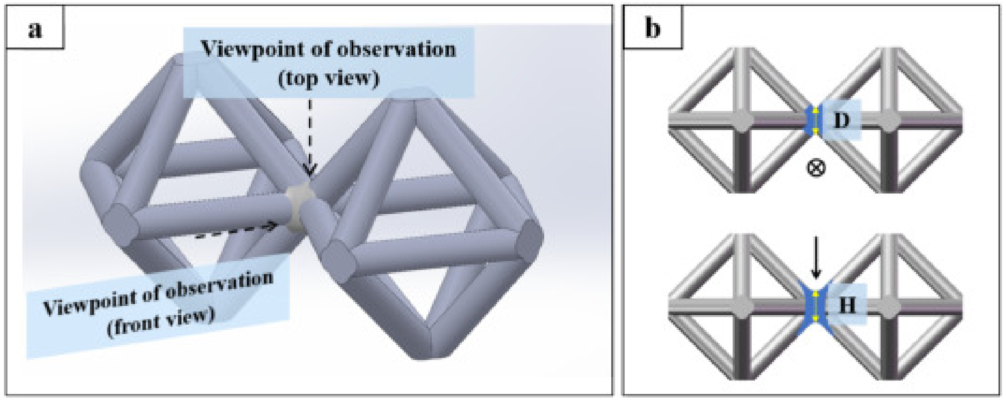

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Fig. 10. Schematic of the observation perspectives for evaluating joint geometric consistency. (a) 3D representation indicating the top-view and front-view angles; (b) corresponding 2D cross-sectional planes used for quantifying weld diameter (D) and penetration depth (H).

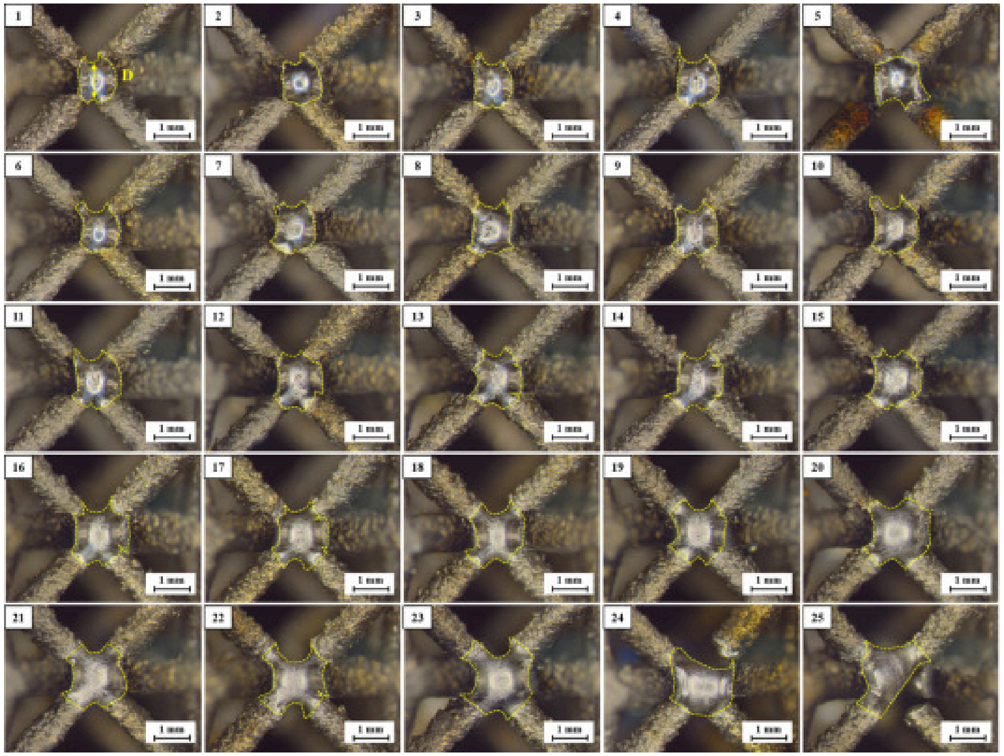

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Fig. 11. Top-view macroscopic morphology of the 25 experimental groups across the 5×5 parameter matrix, illustrating the evolution of fusion diameter (D) as a function of laser power and scanning speed. (The matrix covers the transition from insufficient wetting at low heat input to stable fusion at intermediate heat input and over-melted morphology at excessive heat input.)

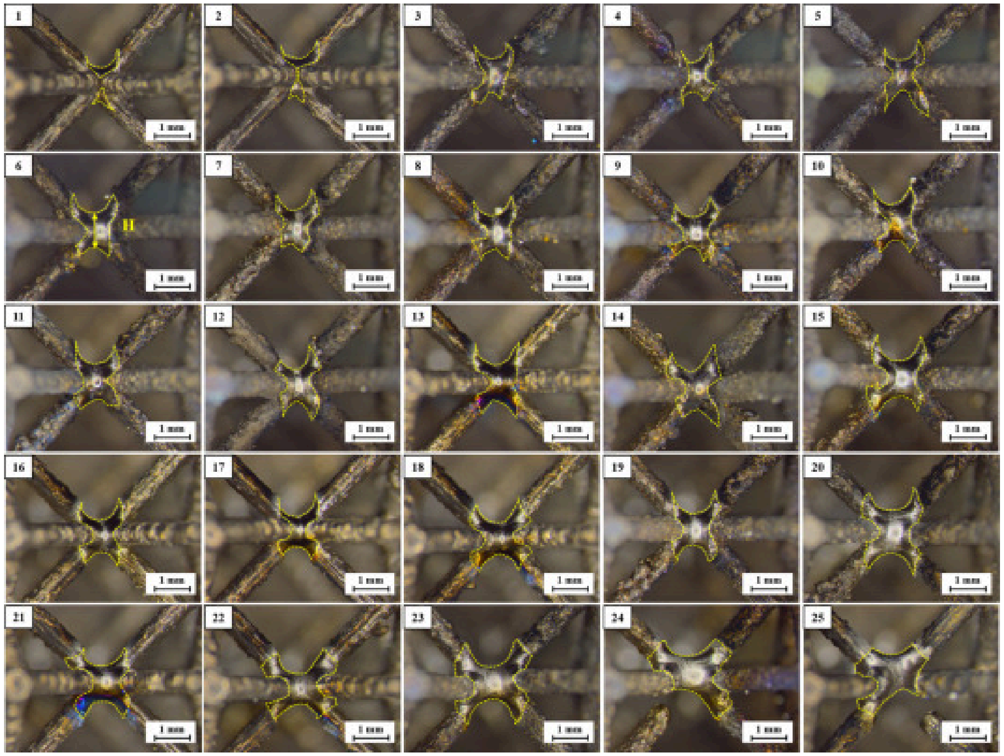

Fig. 12. Front-view macroscopic morphology of the 25 experimental groups across the 5×5 parameter matrix, illustrating the stability of penetration depth (H) and the structural integrity of the lattice ligaments. (Groups 24 and 25 correspond to excessive heat input and are classified as over-melted or failed conditions due to ligament thinning, local collapse, or structural discontinuity.)

Top-view micrographs (Fig. 11) reveal a clear parameter-dependent evolution of the fusion footprint across the 25 experimental groups. According to the power–speed matrix in Table 2, Groups 1–5 correspond to the highest scanning speed of 175 mm/s, whereas Groups 21–25 correspond to the lowest scanning speed of 75 mm/s. Under high-speed or low-power conditions, such as Groups 1, 2, 6, and 11, the fusion footprint remains small and partially discontinuous, indicating insufficient interfacial wetting and limited lateral spreading. With increasing laser power or decreasing scanning speed, the morphology evolves into a continuous and symmetric bead, as represented by Groups 8 and 13, which define the feasible morphology window. In contrast, excessive heat input causes over-expansion, surface sagging, and geometric discontinuity. This is most evident in Groups 24 and 25, corresponding to 800W–75mm/s and 900W–75mm/s, respectively; these conditions are therefore classified as over-melted or failed joints that define the upper boundary of the process window.

Front-view observations (Fig. 12) further clarify the penetration behavior and ligament integrity, complementing the surface fusion-footprint analysis in Fig. 11. In the low-heat-input regime, shallow and spatially non-uniform penetration results in insufficient bonding across the node thickness, especially under high-speed or low-power conditions. This indicates that the molten pool cannot maintain enough lifetime or volume to fully bridge the interface and penetrate into the vertical ligaments. In the intermediate regime, the molten pool couples effectively with the slender struts, producing stable penetration while preserving the original ligament profile and load-bearing geometry. At the high-heat-input end, however, excessive penetration is accompanied by ligament thinning, surface sinking, and local collapse. For Groups 24 and 25, the combination of high power and low speed causes severe heat accumulation and structural discontinuity; thus, these samples should be regarded as failed joints rather than acceptable high-penetration cases. Overall, the feasible process window is bounded by lack-of-fusion at the low-energy side and structural collapse at the high-energy side, indicating that penetration depth must be evaluated together with ligament integrity in lattice-node welding.

The parametric dependence of the joint dimensions is summarized in Fig. 13. The 3D bar charts and trend curves indicate that the fusion diameter D is governed by the synergistic effects of laser power (P) and scanning speed (v). Specifically, D increases monotonically from approximately 1.10mm to 1.32mm as P rises from 500W to 900W, reflecting the expansion of the melt pool driven by enhanced linear energy density. Conversely, D decreases markedly from 1.36mm to 1.10mm as v increases from 75mm/s to 175mm/s, a direct consequence of reduced dwell time and localized energy accumulation. Notably, within the feasible morphology window, the penetration depth H remains relatively stable within a narrow range of approximately 0.95–1.05 mm. This suggests that the dual-ring strategy primarily modulates lateral fusion coverage while maintaining a relatively consistent vertical penetration level under stable welding conditions. However, under excessive heat input, particularly in Groups 24 and 25, the apparent penetration is accompanied by ligament thinning or local collapse and should therefore not be interpreted as effective joint formation. This distinction indicates that geometric consistency, rather than penetration depth alone, must be considered when defining the process window for lattice-node welding.

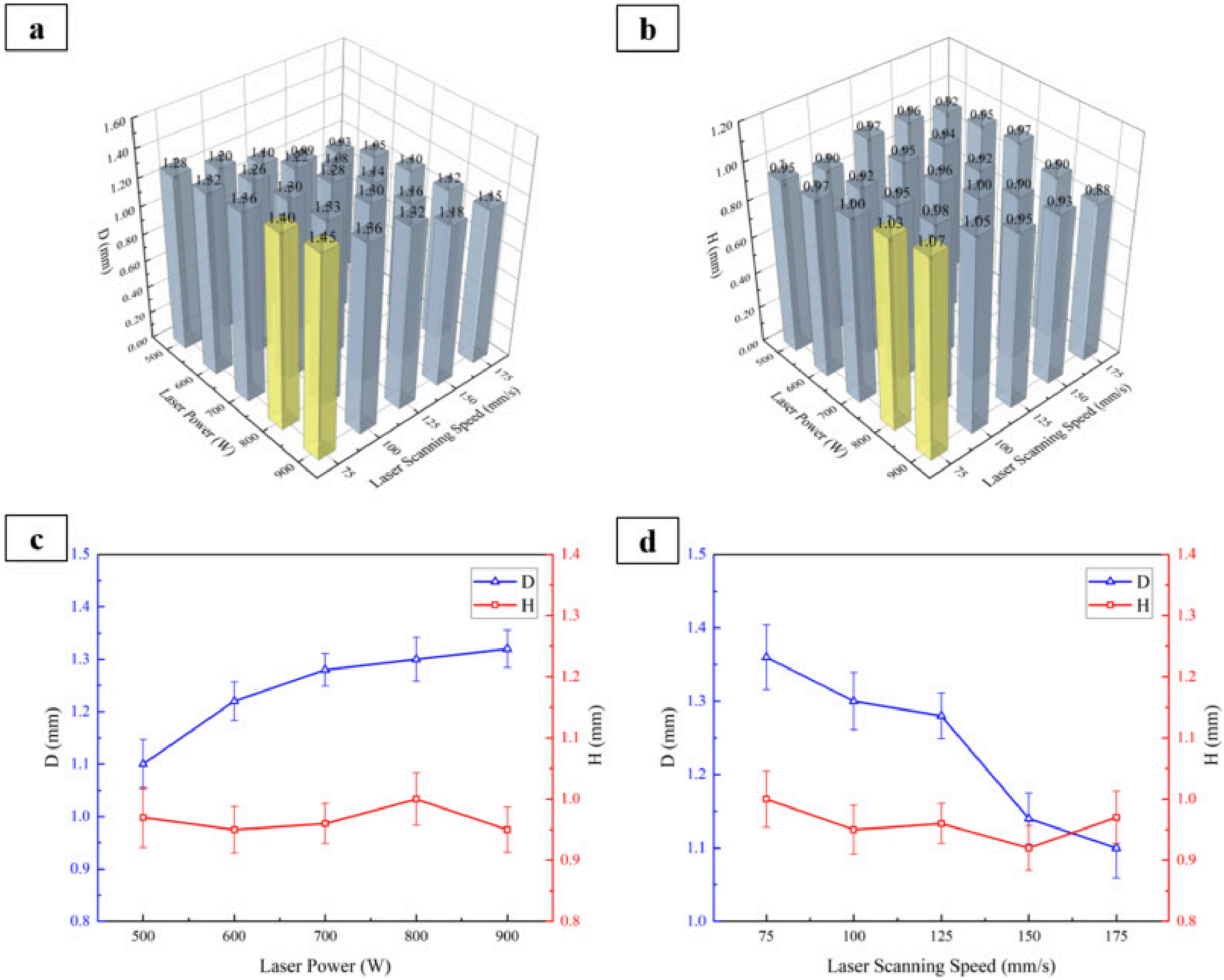

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Fig. 13. Quantitative analysis of the joint geometric dimensions as a function of process parameters. (a, b) 3D bar charts showing the distribution of weld diameter D and penetration depth H across the 5×5 parameter matrix; (c) influence of laser power on D and H at a fixed scanning speed of 125 mm/s; (d) influence of scanning speed on D and H at a fixed laser power of 700 W.

3.2.2. Surface topography and roughness analysis

The surface defect characteristics of the welded lattice nodes were first evaluated using CLSM, with the root-mean-square roughness (*Rq*) employed as a quantitative indicator of surface integrity (Fig. 14, Fig. 15). Representative 3D surface height maps reveal that insufficient or excessive heat input leads to pronounced surface irregularities at the node region. Under low linear energy density conditions, incomplete fusion and discontinuous wetting result in sharp local depressions and uneven height gradients near the node corners. In contrast, excessive heat input promotes excessive molten pool spreading and local surface sagging, which manifests as large-amplitude height fluctuations and elevated roughness values.

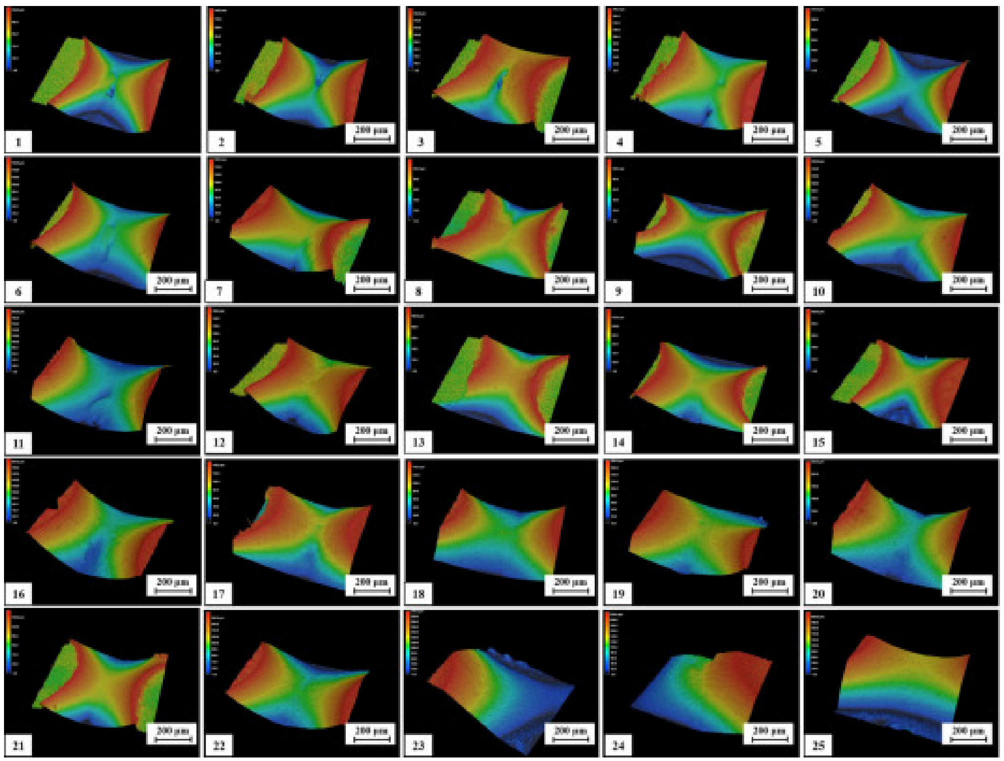

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Fig. 14. 3D surface topography of the welded Ti600 lattice nodes across the 5×5 parameter matrix, obtained via confocal laser scanning microscopy (CLSM).

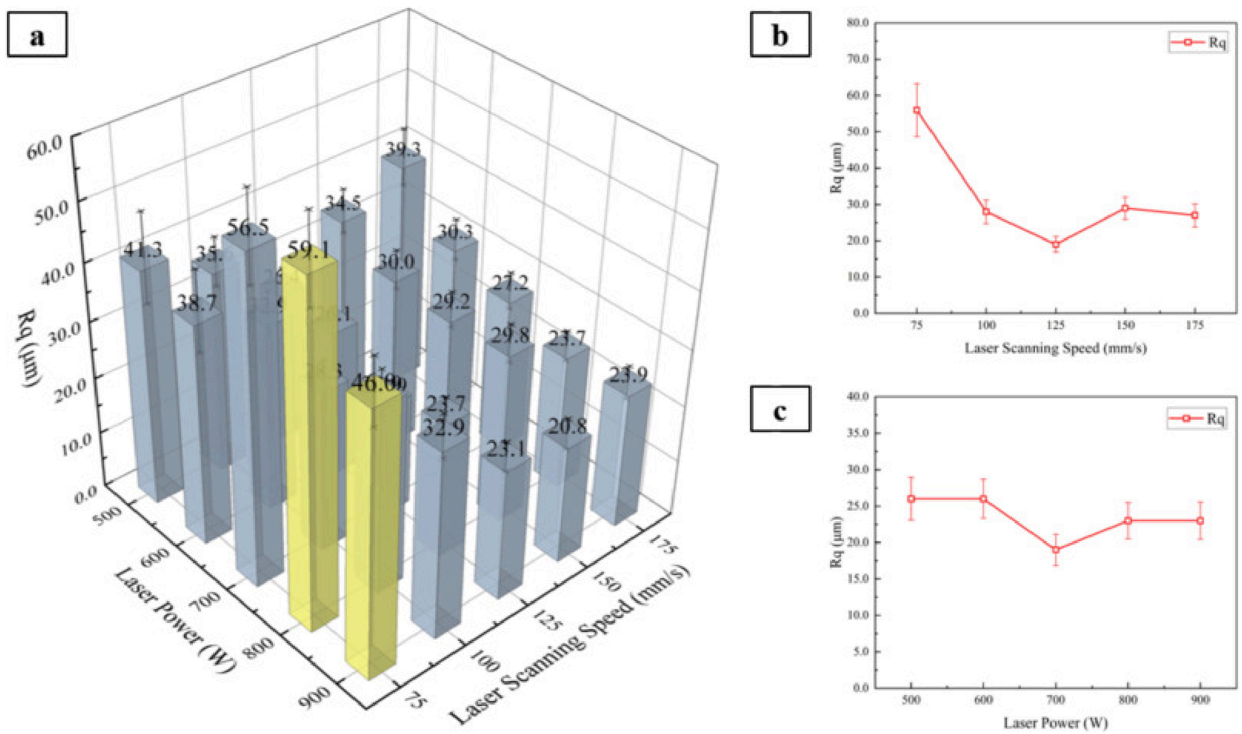

Fig. 15. Statistical analysis of the surface root-mean-square roughness (Rq) as a function of processing parameters. (a) 3D bar chart illustrating the global distribution of Rq; (b) influence of scanning speed on Rq at a fixed laser power of 700W; (c) influence of laser power on Rq at a fixed scanning speed of 125mm/s (Yellow indicates the presence of obvious defects in the welded joint). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The statistical analysis of *Rq* (Fig. 15) demonstrates a clear dependence on both laser scanning speed and laser power. As the scanning speed increases from 75 to 125mm/s, *Rq* decreases significantly, reaching a minimum at intermediate speeds, indicating improved surface smoothness due to stabilized melt pool behavior and reduced thermal accumulation. However, further increases in scanning speed lead to a slight rise in *Rq*, attributed to insufficient energy input and incomplete surface remelting. Similarly, variations in laser power show that moderate power levels yield the lowest roughness, whereas excessive power results in increased *Rq* due to surface collapse and molten metal overflow. These results indicate that surface roughness is governed by the balance between melt pool stability and solidification control, rather than by heat input alone.

To further elucidate the internal defect distribution, X-ray computed tomography (CT) was employed to reconstruct the three-dimensional pore morphology within the welded nodes (Fig. 16). The reconstructed volumes reveal that pores are predominantly concentrated near the node center, with their size, number, and spatial distribution strongly dependent on the processing parameters. Under low heat input conditions (e.g., high scanning speed or low power), a large number of small, irregularly distributed pores are observed, reflecting insufficient molten pool lifetime for gas escape and incomplete fusion at the interface. Conversely, excessive heat input promotes pore coalescence, leading to fewer but significantly larger pores concentrated at the node center.

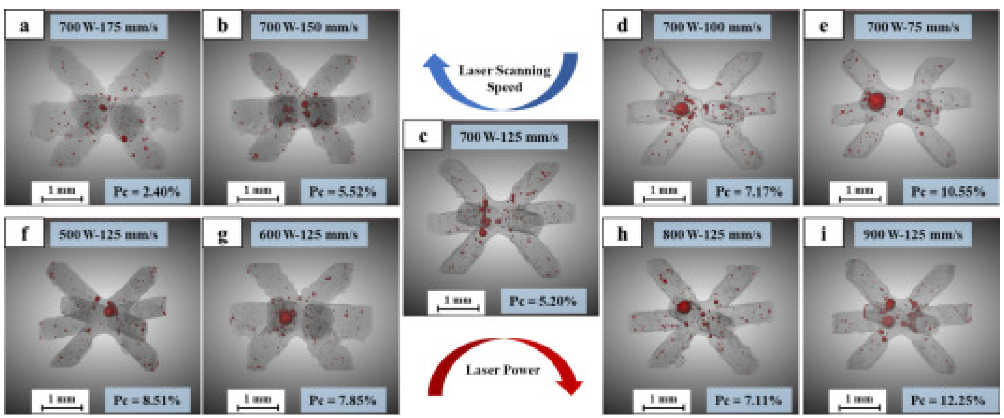

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Fig. 16. X-ray computed tomography (CT) reconstruction of the internal pore morphology and porosity content (Pc) under varying process parameters. (a–e) Evolution of Pc as a function of scanning speed at 700W; (f–i) evolution of Pc as a function of laser power at 125mm/s.

Quantitative analysis of porosity content (*Pc*) confirms the existence of an optimal processing window. At intermediate parameters (e.g., 700W–125mm/s), *Pc* reaches a minimum (~5.20%), corresponding to a stabilized melt pool with sufficient lifetime for bubble rise and escape, as discussed in Section 3.1. In contrast, both insufficient and excessive heat input result in elevated porosity levels. In the former case, rapid solidification traps gas within the melt, while in the latter, strong convection and prolonged molten lifetimes promote pore aggregation rather than elimination.

Taken together, the CLSM and CT results demonstrate that surface roughness and internal porosity are intrinsically coupled through the melt pool dynamics governed by spatiotemporal energy distribution. The dual-ring spiral strategy effectively suppresses defect formation by stabilizing the melt pool and homogenizing the thermal field; however, optimal defect mitigation is achieved only within a narrow processing window where surface smoothness and internal densification are simultaneously maximized. These findings provide a direct morphological and volumetric validation of the energy modulation mechanism proposed in Section 3.1 and establish a solid basis for the mechanical performance analysis presented in the following section.

3.3. Metallographic features and texture evolution

3.3.1. Metallographic comparison between single-ring and dual-ring welds

The correlation between scanning strategy and joint integrity is clearly demonstrated by the comparative metallographic cross-sections and top-view morphologies shown in Fig. 17. Even at the metallographic scale, the path-dependent thermophysical response of the

Ti600 lattice nodes gives rise to pronounced differences in fusion quality and structural continuity. To evaluate the efficacy of the proposed scanning strategy, a comparative analysis was conducted between the single-ring and dual-ring spiral paths. Both strategies were implemented using identical nominal processing parameters (P=700W, v=125mm/s), such that the instantaneous power and linear energy input remained constant. Although the total path length—and thus the cumulative heat input per cycle—differs between the two strategies, this design intentionally isolates the effect of trajectory-induced spatiotemporal energy redistribution on melt pool stability and microstructural evolution, which is the primary focus of this comparison.

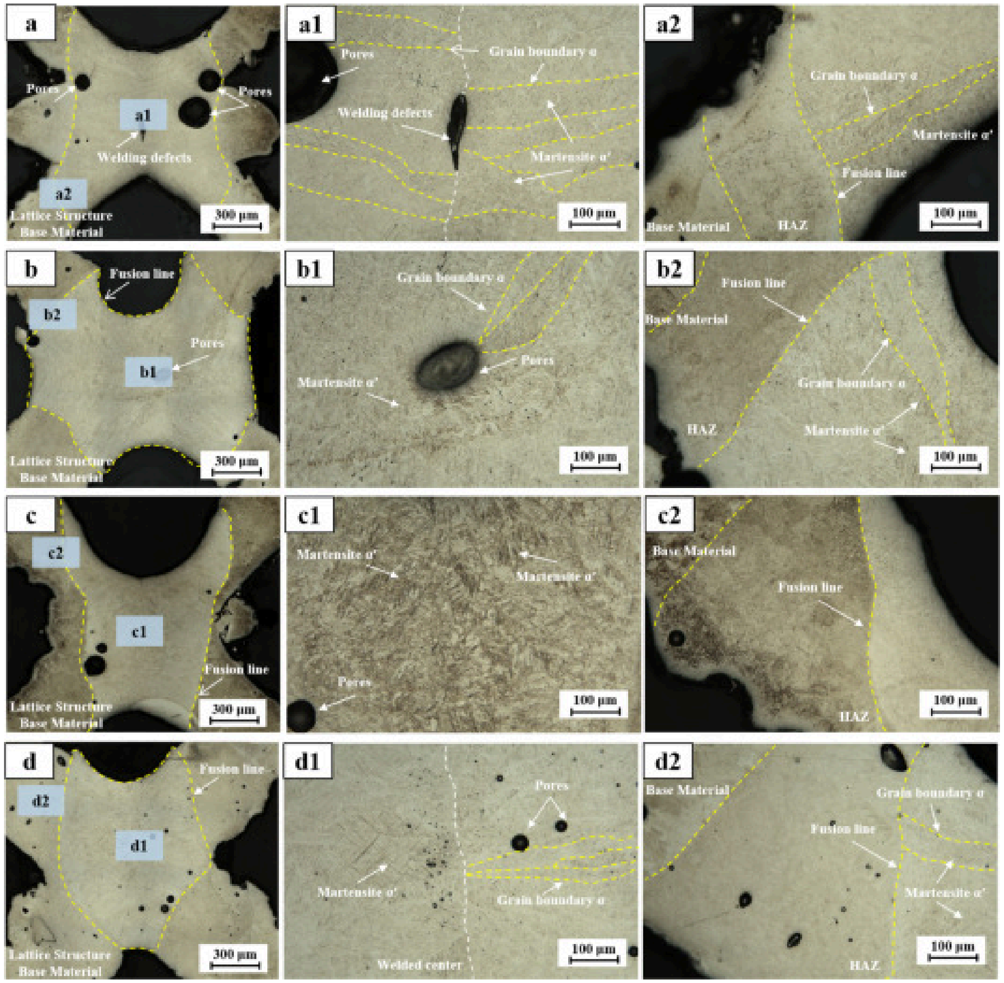

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Fig. 17. Comparative metallographic analysis of Ti600 lattice node joints under different scanning strategies. (a, b) Single-ring scanning results showing: (a) front-view cross-section with a pronounced lack-of-fusion interface at the node center despite locally compact regions, and (b) top-view morphology with irregular and discontinuous fusion boundaries; (c, d) dual-ring spiral scanning results showing: (c) a continuous front-view fusion profile with improved interfacial bonding, and (d) a symmetric, well-defined top-

view fusion footprint. Insets (a1, a2, c1, d2) provide high-magnification details of the α' martensitic microstructure and interfacial fusion characteristics.

For the single-ring scanning condition, the front-view micrograph (Fig. 17a) appears locally compact in some regions; however, the joint quality is dominated by a pronounced interfacial lack-of-fusion defect at the node center rather than by porosity alone. The apparently dense regions correspond mainly to unmelted or insufficiently remelted base material, and therefore should not be interpreted as evidence of continuous metallurgical bonding. The central unbonded interface indicates that the single-ring path is unable to establish a sustained liquid bridge during the initial irradiation stage, as the localized energy input is rapidly extracted by the discrete lattice struts acting as axial heat sinks. Higher-magnification observations (Fig. 17a1 and a2) further reveal a heterogeneous microstructural distribution consisting of coarse acicular α' martensite interspersed with residual grain-boundary, suggesting a strongly non-uniform thermal field and localized solidification imbalance. The corresponding top-view morphology (Fig. 17b) corroborates this instability, exhibiting an irregular fusion boundary accompanied by surface-connected pores, which collectively indicate intermittent interfacial wetting and unstable melt pool behavior under non-modulated energy delivery.

In contrast, the dual-ring spiral scanning strategy yields a substantially improved metallurgical condition, as shown in Fig. 17c and d. Although isolated small pores or local contrast variations may still be observed, the front-view cross-section (Fig. 17c) exhibits a continuous and geometrically uniform fusion profile, with the elimination of the central lack-of-fusion defect and the establishment of metallurgical bonding across the node thickness. Therefore, the improvement in Fig. 17(c,d) should be understood primarily in terms of interfacial continuity, fusion-zone symmetry, and bonding integrity, rather than solely in terms of apparent pore density. The resulting weld microstructure is dominated by refined and interwoven α' martensite laths (Fig. 17c1), reflecting a more homogeneous thermal cycle enabled by the staged energy input and controlled cooling inherent to the dual-ring path. Consistently, the top-view morphology (Fig. 17d) displays a smooth and symmetric fusion boundary with markedly reduced porosity, demonstrating the effectiveness of the geometry-aware scanning strategy in balancing localized heat input and heat extraction along the slender lattice ligaments.

These metallographic observations demonstrate that the single-ring scanning strategy is unable to provide the stable thermal environment required for complete fusion under the severe geometric constraints of lattice nodes. As a result, the microstructural features of single-ring joints are dominated by process-induced defects rather than intrinsic solidification characteristics. Accordingly, the subsequent in-depth crystallographic analysis using Electron Backscatter Diffraction (EBSD) is focused on the dual-ring joints, which represent the stabilized and technologically relevant processing condition for elucidating spatially resolved grain orientation, texture evolution, and phase transformation behavior.

3.3.2. Prior-β grain reconstruction and growth directionality

The reconstruction of the prior-β grain structure from the room-temperature hexagonal close-packed (HCP) α' phase is predicated on the strict crystallographic Burgers Orientation Relationship (BOR). During the rapid cooling inherent to laser welding of the Ti600 alloy, the parent body-centered cubic (BCC) β phase transforms into the metastable α' martensite phase through a diffusionless shear mechanism. This transformation follows a specific atomic mapping between the two lattices: $\{110\}_{\beta} /\!/ (0001)_{\alpha'}$ and $\langle 111\rangle_{\beta} /\!/ <11\bar{2}0>_{\alpha'}$. Due to the high symmetry of the BCC β phase, a single parent grain can theoretically generate 12 crystallographically equivalent α' variants during the β to α' transformation. In the EBSD-based reconstruction process, the unique orientation of the parent β phase is mathematically back-calculated by identifying the optimal parent orientation that minimizes angular deviation across multiple variants within a localized cluster. This methodology effectively reveals the high-temperature solidification structure, allowing for a direct assessment of how the dual-ring spiral strategy disrupts typical epitaxial growth patterns.

The reconstructed prior-β grain morphologies for the single-ring and dual-ring conditions are presented in Fig. 18. The EBSD reconstruction regions correspond to the same front-view and top-view observation locations used for the metallographic comparison in Fig. 17, covering the representative fusion zone and adjacent interfacial regions of the welded node. Under identical linear energy density conditions, the reconstructed prior-β grain morphologies for the single-ring and dual-ring conditions exhibit contrasting growth kinetics. Under the single-ring strategy, the front-view reconstruction (Fig. 18a) is dominated by coarse columnar grains growing epitaxially from the fusion boundary toward the weld center. These grains are strongly aligned with the primary heat extraction direction along the lattice struts, which act as efficient axial heat sinks. In addition, the presence of large spherical pores (white regions) suggests unstable melt pool dynamics that locally interrupted grain continuity and entrapped vapor during solidification. In the top-view reconstruction (Fig. 18b), pronounced radial grain impingement toward the node center is observed. Such centerline impingement is characteristic of solidification under steep thermal gradients and limited lateral heat redistribution, and is commonly associated with localized microstructural heterogeneity.

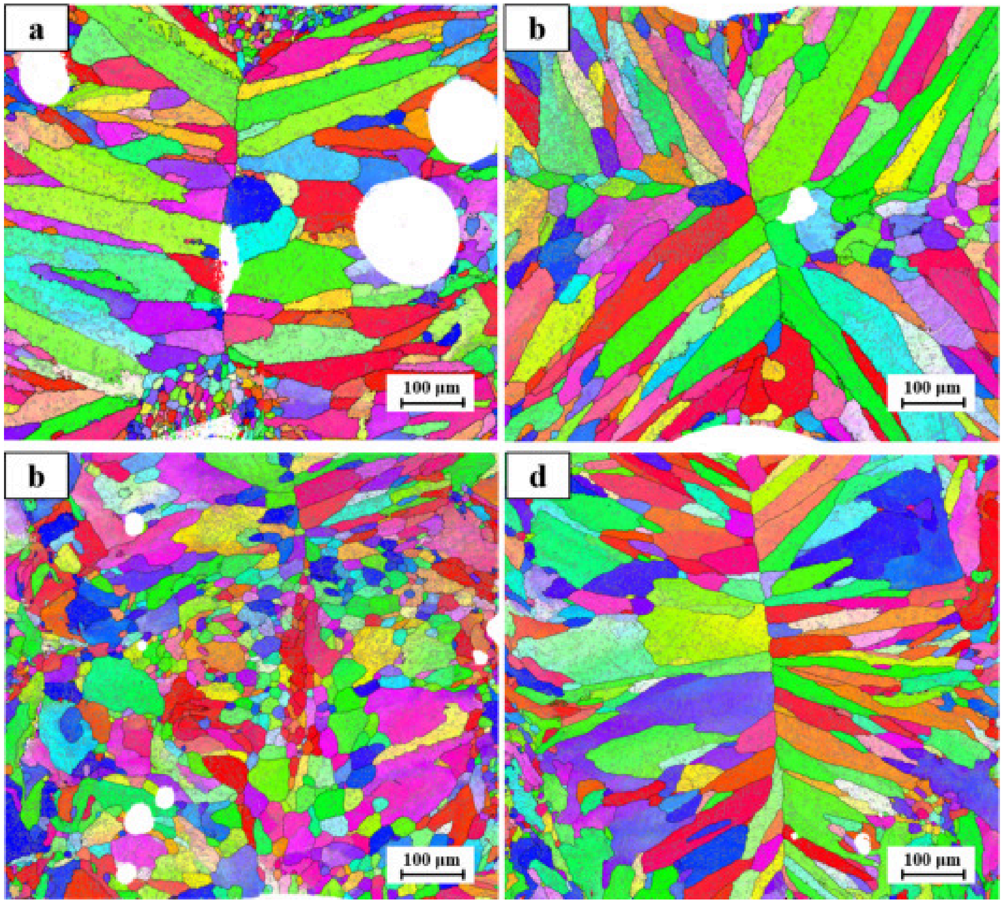

Fig. 18. Prior-β grain reconstruction of Ti600 lattice node joints derived via the Burgers Orientation Relationship (BOR), with EBSD scan regions corresponding to the front-view and top-view observation locations shown in Fig. 17. (a) Single-ring front-view showing coarse epitaxial columnar grains and entrapped porosity; (b) single-ring top-view showing prominent radial grain impingement; (c) dual-ring spiral front-view exhibiting grain refinement and disrupted epitaxial growth; (d) dual-ring spiral top-view showing a homogenized grain distribution with reduced anisotropy.

In contrast, the dual-ring spiral strategy markedly alters the grain growth behavior through its staged energy input. The front-view reconstruction (Fig. 18c) exhibits refined prior-β grains with disrupted epitaxial growth and the appearance of equiaxed grain clusters in the central fusion region. Because the local thermal gradient and solidification rate were not directly measured, this evolution should not be regarded as quantitative evidence of a specific reduction in G or G/R. Instead, it suggests that the spiral trajectory weakens persistent directional growth and promotes a more spatially distributed solidification condition, which is consistent with the observed partial columnar-to-equiaxed transition. The top-view reconstruction (Fig. 18d) further reveals a more uniform grain distribution with suppressed centerline impingement, supporting the interpretation that trajectory-induced energy redistribution can reduce microstructural directionality.

Overall, the dual-ring spiral strategy not only mitigates macroscopic defects but also appears to shift the solidification morphology from strongly directional columnar growth toward a more spatially distributed grain-growth mode. It should be noted that the prior-β grain reconstruction in Fig. 18 is used here to support a qualitative comparison of grain-growth directionality between the single-ring and dual-ring strategies, rather than to provide a quantitative texture-intensity comparison between the two trajectories. By matching the laser scanning path to the intrinsic heat dissipation characteristics of the Ti600 lattice, this strategy establishes a robust microstructural foundation for the improved joint performance discussed in the following section.

3.3.3. Localized microstructural features and crystallographic texture evolution

To elucidate the three-dimensional crystallographic gradients induced by the dual-ring spiral strategy, localized EBSD analyses were conducted on the optimized joints along two orthogonal planes, as presented in Fig. 19 and Fig. 20. This spatially-resolved characterization focused on the heat-affected zone (HAZ), the weld center, and the fusion boundaries (bottom/edge), enabling a comprehensive assessment of how spiral energy delivery modulates the α’ martensitic microstructure in lattice-confined welds. As shown in the front-view (Fig. 19a) and top-view (Fig. 20a) observations, the HAZ exhibits a relatively uniform α’ lath thickness of approximately 4.38 μm in both front-view and top-view observations, indicating a stable and repeatable thermal response of the LPBF Ti600 ligaments under sub-solidus heating. Despite the thermal exposure, this region retains a high pole figure (PF) intensity of up to 17.84, suggesting that the fiber texture inherited from the additive manufacturing process is largely preserved in the absence of complete remelting. This sub-region acts as a structural transition, maintaining the crystallographic “memory” of the base material while accommodating the thermal stresses of the welding process.

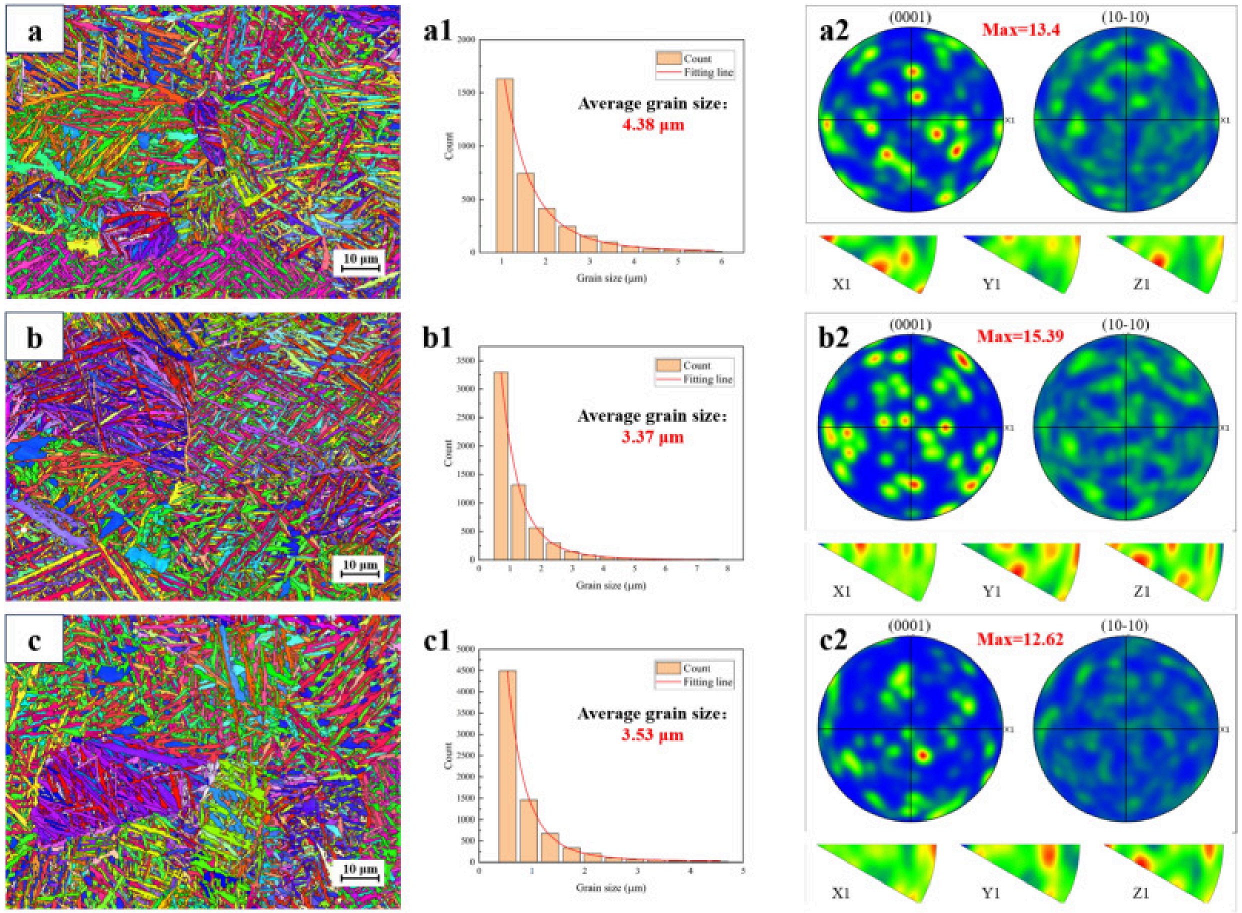

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Fig. 19. Localized EBSD analysis and grain size statistics of the optimized dual-ring joint (Front-view). (a) HAZ region showing inherited LPBF texture; (b) weld center exhibiting the finest α' laths; (c) weld bottom showing transitional solidification features.

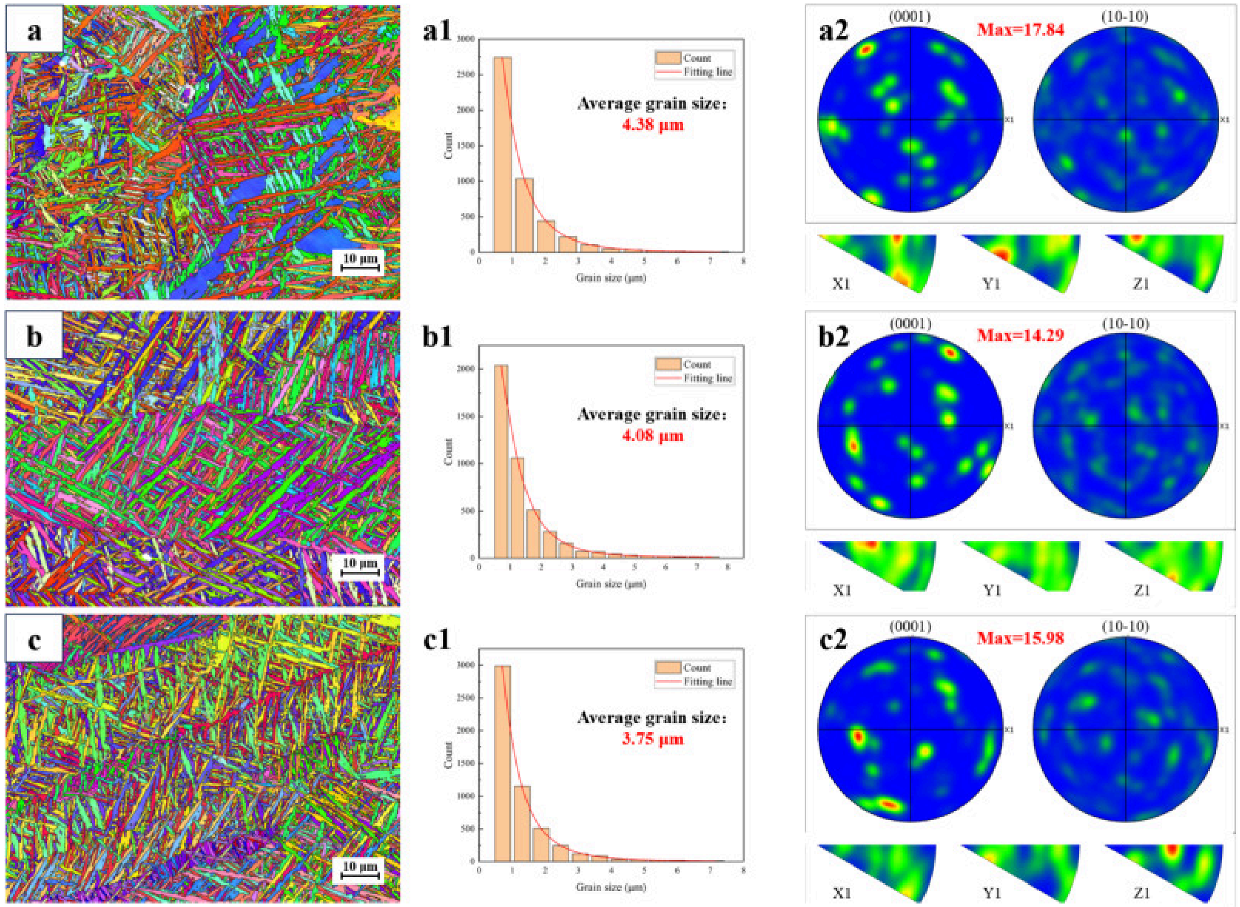

Fig. 20. Localized EBSD analysis and grain size statistics of the optimized dual-ring joint (Top-view). (a) HAZ region showing high PF intensity; (b) weld center showing randomized texture; (c) weld edge illustrating the transition between the fusion zone and base strut.

In contrast, the weld center displays pronounced microstructural refinement with a clear dependency on the observation plane. The front-view section (Fig. 19b) exhibits the finest martensitic laths, with an average thickness of 3.37 μm, whereas the corresponding top-view region (Fig. 20b) displays slightly coarser laths of approximately 4.08 μm. This anisotropy arises from the three-dimensional heat sink effect intrinsic to the lattice geometry. The front-view (vertical section) captures the mid-thickness of the node where heat is simultaneously extracted through four converging struts, creating a multi-directional thermal drain that maximizes the cooling rate. The top-view, by contrast, reflects surface-dominated solidification where radiative and convective losses are comparatively less efficient than the conductive cooling through the solid ligaments.

Beyond grain refinement, localized EBSD results reveal a spatial variation in crystallographic texture within the optimized dual-ring joint. The HAZ retains a relatively high pole figure intensity of approximately 17.84, reflecting the crystallographic memory inherited from the LPBF base material under sub-solidus thermal exposure. In contrast, the weld center exhibits a lower local pole figure intensity of approximately 14.29, indicating reduced crystallographic directionality within the remelted fusion zone of the same optimized joint. This local decrease should therefore be interpreted as an intra-joint texture variation between the HAZ and weld center, rather than as direct quantitative evidence of texture weakening from the single-ring to the dual-ring strategy. Nevertheless, this observation is consistent with the prior-β reconstruction results, which suggest that the spiral trajectory disrupts persistent directional grain growth and promotes a more spatially distributed solidification morphology. At the fusion boundaries, including the bottom (Fig. 19c) and edge regions (Fig. 20c), the average lath thickness ranges between 3.53 and 3.75 μm, representing a smooth microstructural “buffer” between the refined weld core and the thermally affected ligaments. This gradual variation in grain morphology ensures metallurgical continuity and local structural compatibility, effectively bridging the crystallographic gap between the discrete LPBF members.

3.4. Mechanical performance and fracture behavior

To accurately assess the load-bearing capacity of the welded Ti600 lattice nodes, a customized uniaxial tensile testing configuration was developed, as schematically illustrated in Fig. 21. Owing to the complex topology and slender geometry of the lattice ligaments, a dedicated stretching fixture was designed to ensure coaxial alignment and to suppress parasitic bending moments during loading. The fixture was manufactured from high-strength steel by conventional machining, and consisted of two clamping blocks with matching positioning grooves to constrain the pyramid-like lattice units. The lattice specimens were rigidly fixed in the grooves by mechanical clamping, enabling the applied tensile force to be transmitted directly through the welded node. Under this configuration, the welded joint constitutes the critical load-bearing section along the force path, ensuring that the measured mechanical response mainly reflects the metallurgical integrity of the joint rather than premature failure caused by fixture misalignment or surrounding-ligament bending.

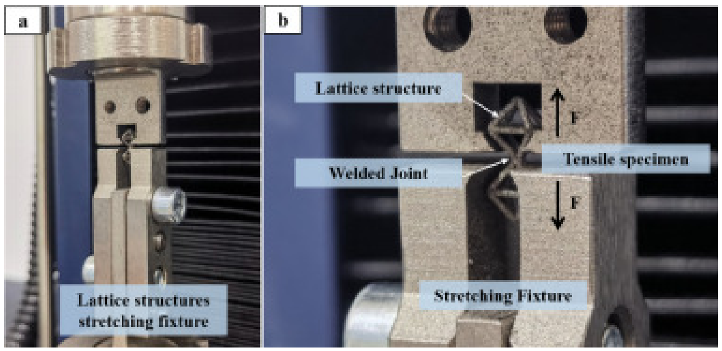

Fig. 21. Experimental configuration for single-node tensile testing. (a) Overview of the machined high-strength-steel stretching fixture used for positioning and clamping the lattice specimen; (b) detailed view of the lattice structure alignment and tensile loading direction.

Based on this testing methodology, quasi-static tensile tests were conducted at a crosshead speed of 0.5 mm/min, and the maximum tensile force (Fmax) was systematically evaluated across the 5×5 processing matrix to establish a quantitative correlation between mechanical performance and the spatiotemporal energy modulation mechanisms discussed in 3.1 Dual-ring spiral path principle and melt pool dynamics, 3.2 Macro-morphology and defect distribution, 3.3 Metallographic features and texture evolution (Fig. 22). The measured values of F_{max} reveal a well-defined processing window in which the welded joints exhibit the highest load-bearing capacity. At a constant laser power of 700W, F_{max} increases markedly as the scanning speed rises from 75 to 150mm/s, reaching a peak value of 452. 4N. This enhancement is attributed to the suppression of excessive thermal accumulation and the establishment of a stabilized melt pool, which collectively promote defect-free fusion and uniform microstructural refinement. However, further increasing the scanning speed beyond 150mm/s results in a sharp degradation of tensile strength due to insufficient energy input, leading to incomplete interfacial bonding and lack-of-fusion defects.

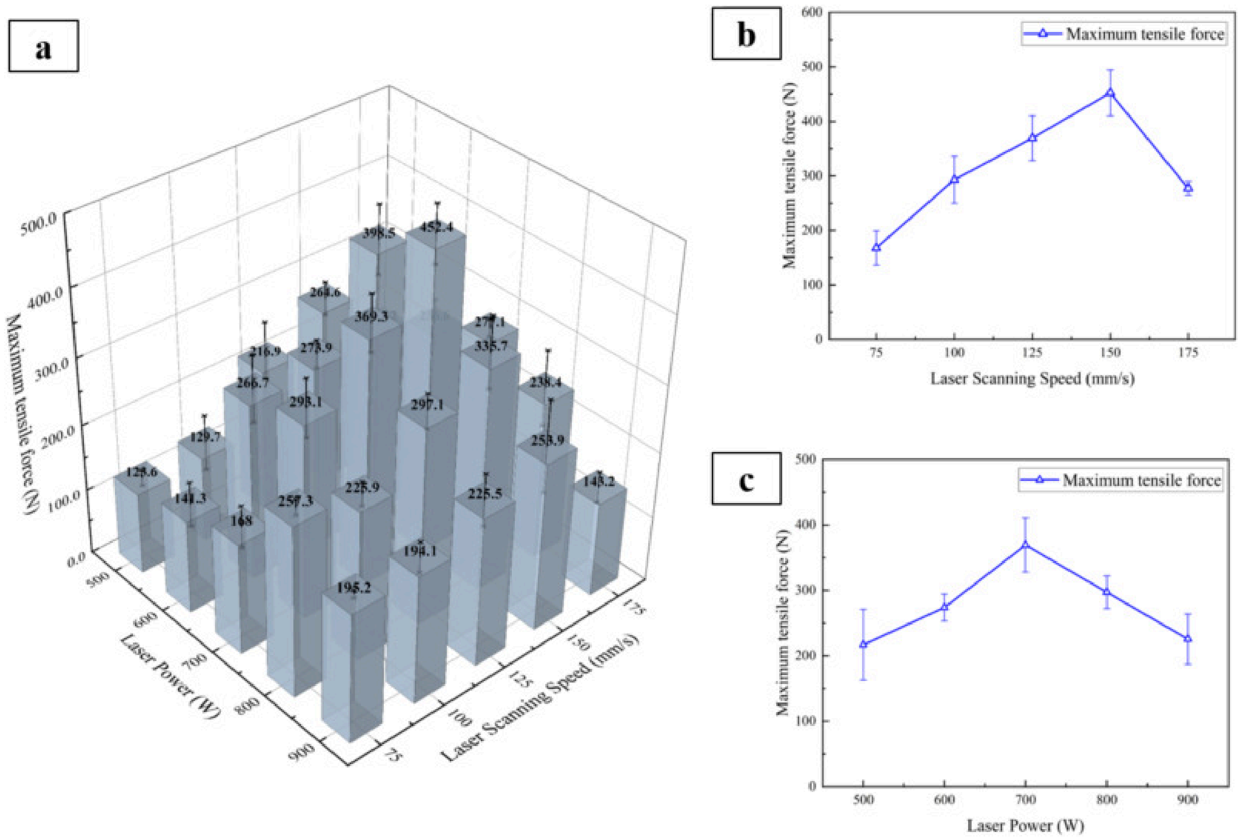

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Fig. 22. Maximum tensile force of the welded Ti600 nodes across the parameter space. (a) 3D statistical bar chart of Fmax as a function of power and speed; (b) influence of scanning speed at 700W; (c) influence of laser power at 150mm/s.

A similar non-monotonic dependence is observed with respect to laser power. The maximum tensile force is achieved at an intermediate power level of 700W, whereas both lower and higher powers result in reduced joint performance. At suboptimal power levels, insufficient penetration and interfacial discontinuities act as dominant stress concentrators. In contrast, excessive laser power induces intense vapor recoil pressure and melt pool instability, promoting surface sagging and internal porosity that diminish the effective load-bearing cross-section of the node. These results demonstrate that the mechanical performance of the welded lattice joints is maximized only within a narrow processing window, where melt pool stability, surface smoothness, and internal densification are simultaneously optimized by the dual-ring spiral scanning strategy.

It should be noted that the present mechanical evaluation focuses on the maximum node-level load-bearing capacity and the corresponding fracture mode, rather than on a full constitutive description of the welded lattice joint. Because the single-node specimen has a discrete ligament-supported load path, the load–displacement response can be influenced by local ligament bending, fixture seating, and variations in fracture location. Therefore, F_{max} is used as a comparative indicator of joint load-bearing capacity and is interpreted together with macro-morphology, CT porosity, and SEM fractography. From an application perspective, the welded node may also be subjected to bending, shear, compression, or cyclic loading, under which defect location, ligament thinning, and lack-of-fusion may affect the local failure response differently from uniaxial tension.

The fracture morphologies of the welded Ti600 lattice joints under different processing conditions are summarized in Fig. 23, with representative high-magnification features shown in Fig. 24. Together, these figures elucidate how processing-induced defects and local microstructural states govern the tensile failure behavior.

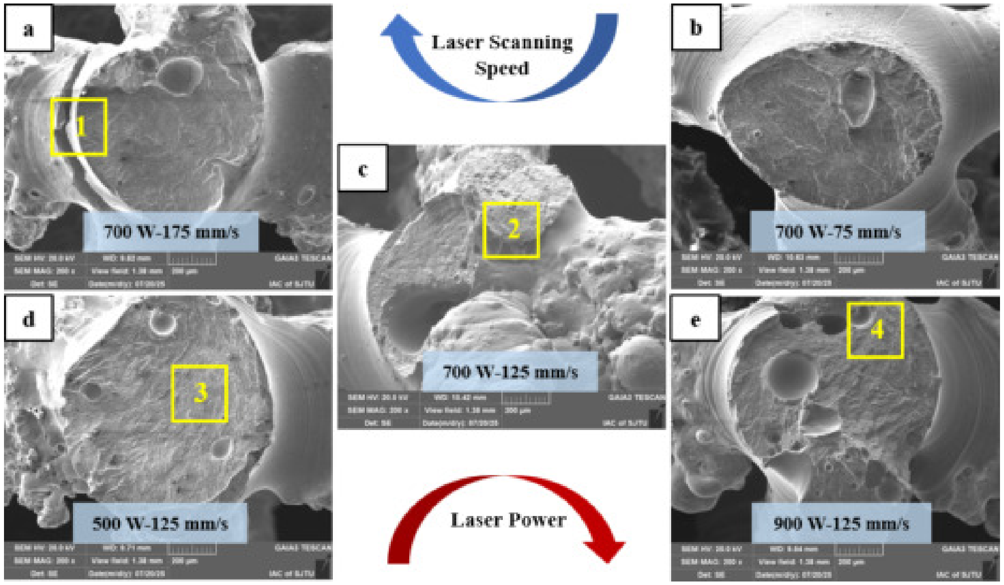

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Fig. 23. SEM macro-morphologies of fracture surfaces of welded Ti600 lattice joints under different processing parameters. (a) 700W–175mm/s, lack-of-fusion–dominated fracture; (b) 700W–75mm/s, porosity-dominated fracture; (c) 700W–125mm/s, optimized condition with fracture occurring in the HAZ; (d) 500W–125mm/s, lack-of-fusion–assisted fracture; (e) 900W–125mm/s, porosity-assisted fracture.

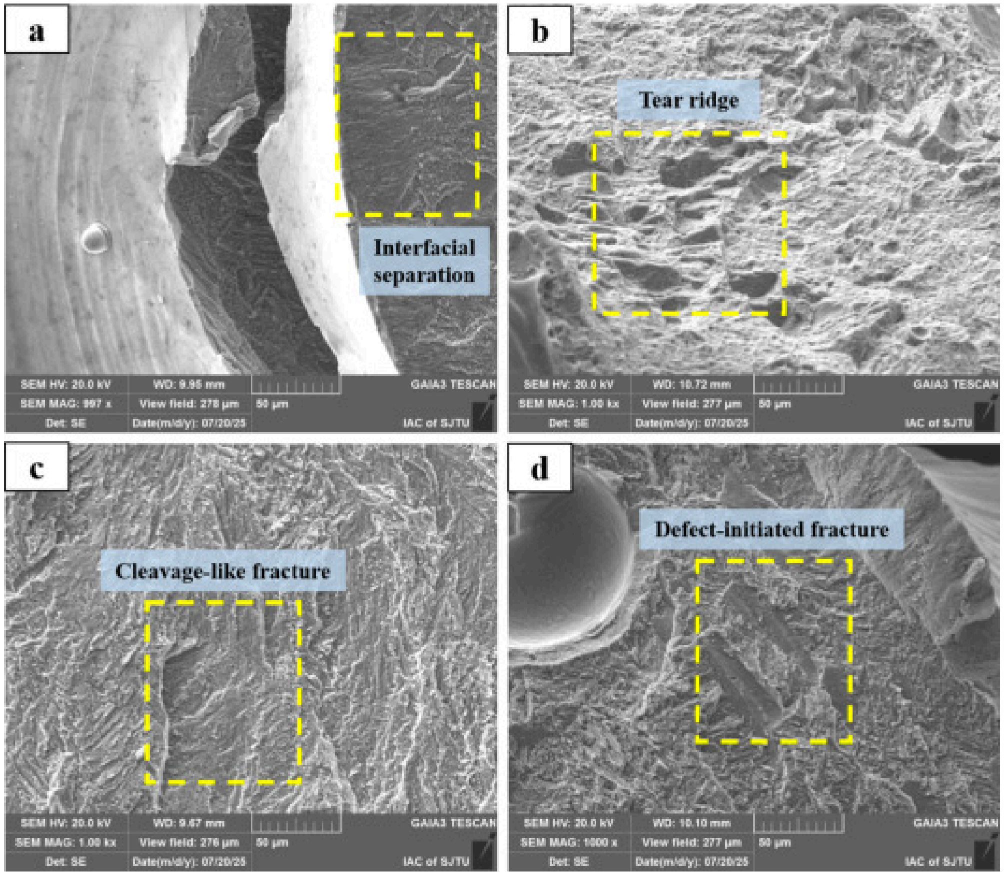

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Fig. 24. High-magnification SEM images of representative fracture features corresponding to the marked regions in Fig. 23. (a) Region 1 showing interfacial separation with unbonded facets; (b) Region 2 showing a dense ductile dimple morphology under the optimized condition; (c) Region 3 showing lack-of-fusion features accompanied by micro-pores; (d) Region 4 showing large gas pores associated with defect-initiated fracture.

Under low heat input conditions (e.g., 700W–175mm/s and 500W–125mm/s), fracture is dominated by interfacial discontinuities. As shown in Fig. 24a (Region 1 in Fig. 23a), the fracture surface exhibits interfacial separation with locally planar facets and limited plastic deformation, characteristic of lack-of-fusion–controlled failure caused by incomplete liquid bridging. Cleavage-like or quasi-cleavage features are also observed (Fig. 24c, Region 3), further confirming a defect-dominated fracture mode under insufficient energy input.

In contrast, excessive heat input (e.g., 700W–75mm/s and 900W–125mm/s) leads to porosity-controlled fracture. Fig. 24d (Region 4) reveals crack initiation adjacent to large, smooth-walled pores formed by melt-pool instability and keyhole collapse, which substantially reduce the effective load-bearing area. This behavior is consistent with the elevated porosity levels quantified by CT analysis in Section 3.2.2.

A fundamentally different failure mechanism is observed under the optimized condition (700W–125mm/s). As shown in Fig. 24b (Region 2), the fracture surface is characterized by ductile tearing with well-developed tear ridges, indicating significant local plastic deformation. Consistent with the macroscopic fracture location in Fig. 23c, failure occurs in the HAZ rather than within the fusion zone, demonstrating that the weld region exhibits higher strength than the surrounding lattice ligaments. This transition reflects the combined effects of defect suppression and a more homogeneous microstructural response achieved through optimized dual-ring spiral energy delivery.

Overall, Fig. 23, Fig. 24 reveal a clear transition from defect-assisted brittle or cleavage-like fracture to microstructure-controlled ductile tearing as the processing parameters approach the optimized regime. Insufficient or excessive heat input promotes lack-of-fusion or porosity-governed failure, whereas optimized spatiotemporal energy modulation stabilizes the melt pool and shifts the fracture locus away from the weld center toward the HAZ, confirming that the welded joint is no longer the mechanically limiting region.

4. Conclusion

In this study, the weldability of additively manufactured Ti600 high-temperature titanium alloy lattice structures was systematically investigated using a geometry-aware, dual-ring spiral laser scanning strategy. By tailoring the scan trajectory to the intrinsic heat dissipation characteristics of lattice nodes, the coupling between path-controlled heat input, melt-pool behavior, microstructural evolution, and joint performance was elucidated over a wide range of laser power (500–900 W) and scanning speed (75–175 mm/s).The principal findings can be summarized as follows.

(1) Macro-morphology and defect control: The fusion footprint and internal soundness of the welded nodes are strongly governed by the spatiotemporal distribution of heat input. An intermediate processing window (approximately 700 W and 125–150 mm/s) enables stable melt-pool evolution and effective interfacial bridging, resulting in symmetric fusion zones and minimal internal porosity (~5.20%). This confirms that the dual-ring spiral path decouples surface wetting from penetration control, thereby accommodating the inherent topographic irregularities of LPBF lattice nodes.

(2) Crystallographic and microstructural evolution: The reconstructed prior-β grains reveal a qualitative difference in grain-growth directionality between the single-ring and dual-ring strategies, suggesting that the spiral trajectory disrupts persistent directional growth and promotes a partial columnar-to-equiaxed transition in the fusion zone. Localized EBSD analysis of the optimized dual-ring joint further shows that the pole figure intensity decreases from 17.84 in the heat-affected region to 14.29 in the weld center, indicating reduced local crystallographic directionality after remelting. This decrease, however, represents an intra-joint texture variation within the optimized dual-ring weld and should not be interpreted as direct quantitative evidence of texture weakening relative to the single-ring trajectory.

(3) Mechanical performance and failure behavior: Tensile testing demonstrates that optimized parameters (e.g., 700W and 150mm/s) achieve the highest node-level load-bearing capacity, with a maximum tensile force of 452.4N. A clear transition in failure mode is observed, from defect-assisted brittle fracture under insufficient or excessive heat input to microstructure-controlled ductile failure within the optimized regime. Notably, fracture shifts from the weld center to the heat-affected zone (HAZ), indicating that the metallurgical integrity of the welded joint exceeds that of the surrounding lattice ligaments and that the weld is no longer the mechanically limiting region.

Beyond these specific results, this work establishes a spatiotemporal energy modulation principle in which the scan trajectory functions as a thermal buffer, reconciling the high energy density required for titanium fusion with the rapid, anisotropic heat dissipation of lattice architectures. The transferable aspect of the proposed strategy lies in the staged energy-delivery concept of inner-ring interfacial bridging followed by outer-ring coverage and lateral heat redistribution, rather than in a single fixed trajectory geometry. This concept of trajectory-controlled energy redistribution is broadly applicable and provides mechanism-based design guidance for other complex-topology or thin-walled metallic systems. For practical extension, the loop dimensions and overlap ratio should be adjusted according to the characteristic node scale, laser spot size, and material-dependent thermal response, rather than regarded as fixed constants. More specifically, material-dependent factors such as laser absorptivity, thermal diffusivity, melting range, molten-pool viscosity, and oxidation sensitivity should be considered when selecting the power–speed combination and loop overlap. For different lattice geometries, the trajectory scale should be further related to strut diameter, node overlap area, local heat-sink number, and expected interfacial gap. From a practical perspective, the proposed strategy enables repeatable and reliable laser additive-welding of porous metallic structures, reducing reliance on trial-and-error parameter optimization and supporting the scalable assembly of multifunctional aerospace lattice architectures.

CRediT authorship contribution statement

Cheng Liu: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Ming Lou:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Han Yu:** Resources, Methodology, Formal analysis, Data curation. **Zixuan Chen:** Software, Methodology, Investigation. **Bowen Zhang:** Software, Resources, Methodology, Investigation. **Yunwu Ma:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Yongbing Li:** Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used translation software in order to polish language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Acknowledgments

The authors are grateful to the financial supports by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (Grant No. JYB2025XDXM214).

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